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Kim et al.

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(54) **AIR CONDITIONER, AIR CONDITIONER CONTROL METHOD, AND AIR CONDITIONER CONTROL SYSTEM**

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(Continued)

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Primary Examiner — Chun Cao

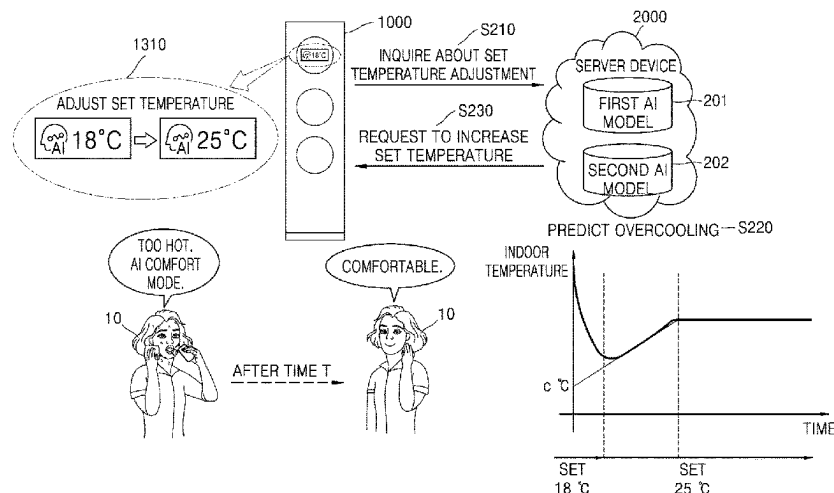
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(57) **ABSTRACT**

An air conditioner includes a communication interface to communicate with a server device, a memory to store one or more instructions, and at least one processor configured to execute the stored one or more instructions to transmit, to the server device through the communication interface, current state information including one or more of operation time information of the air conditioner, set temperature information of the air conditioner, and current indoor temperature information, receive a set temperature increase request corresponding to the transmitted current state information from the server device through the communication interface, and adjust a set temperature of the air conditioner based on the received set temperature increase request, and the set temperature increase request is received when an overcooling period is identified.

17 Claims, 26 Drawing Sheets



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F24F 110/12 (2018.01)
F24F 110/20 (2018.01)
F24F 110/22 (2018.01)
F24F 110/30 (2018.01)

(52) **U.S. Cl.**

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2110/22 (2018.01); *F24F 2110/30* (2018.01);
G05B 2219/2614 (2013.01)

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F24F 11/62; *F24F 11/64*; *F24F 11/72*;
F24F 2130/10; *G05B 19/042*; *G05B*
2219/2614; *G05B 13/0265*; *G05B 15/02*;
G06N 20/00
USPC 700/180

See application file for complete search history.

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FIG. 1
(RELATED ART)

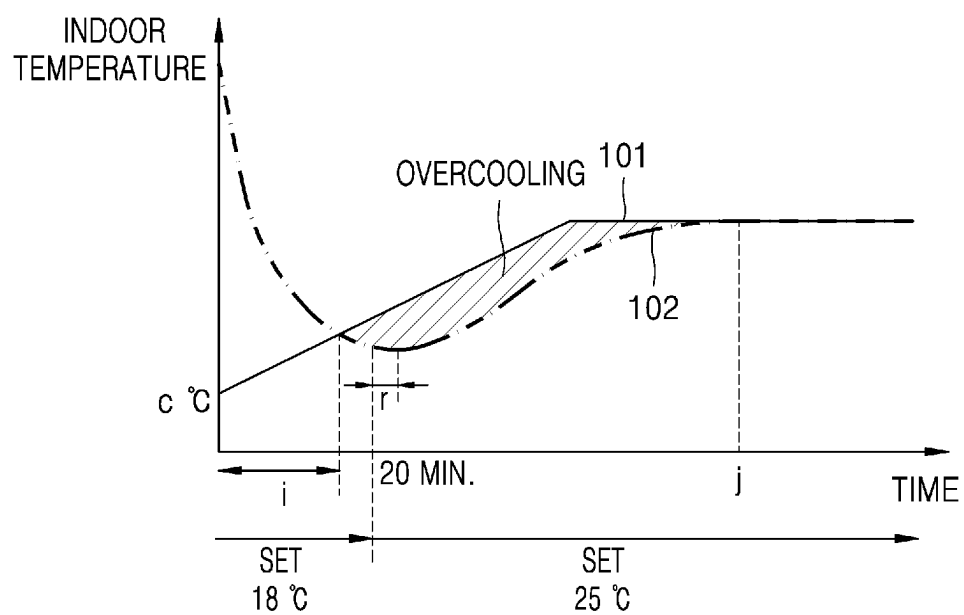
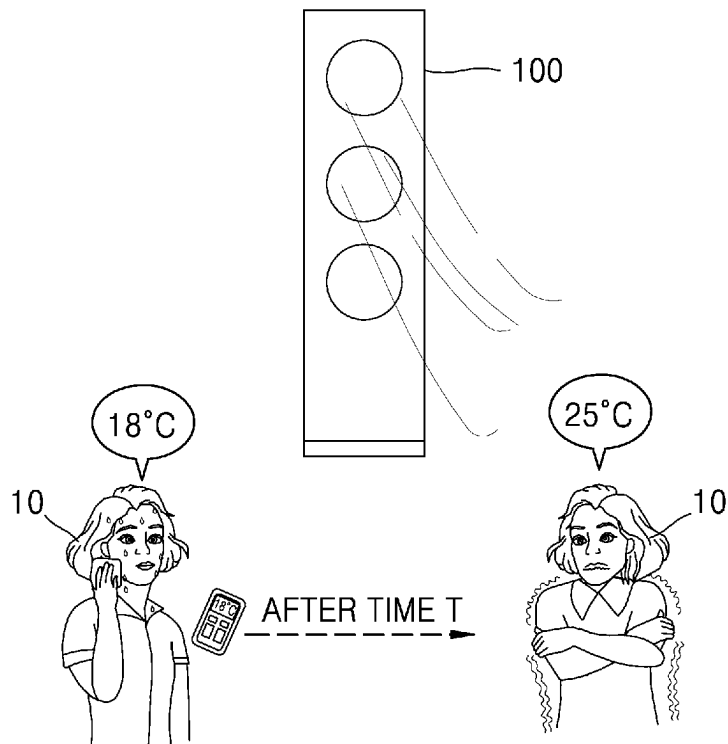


FIG. 2

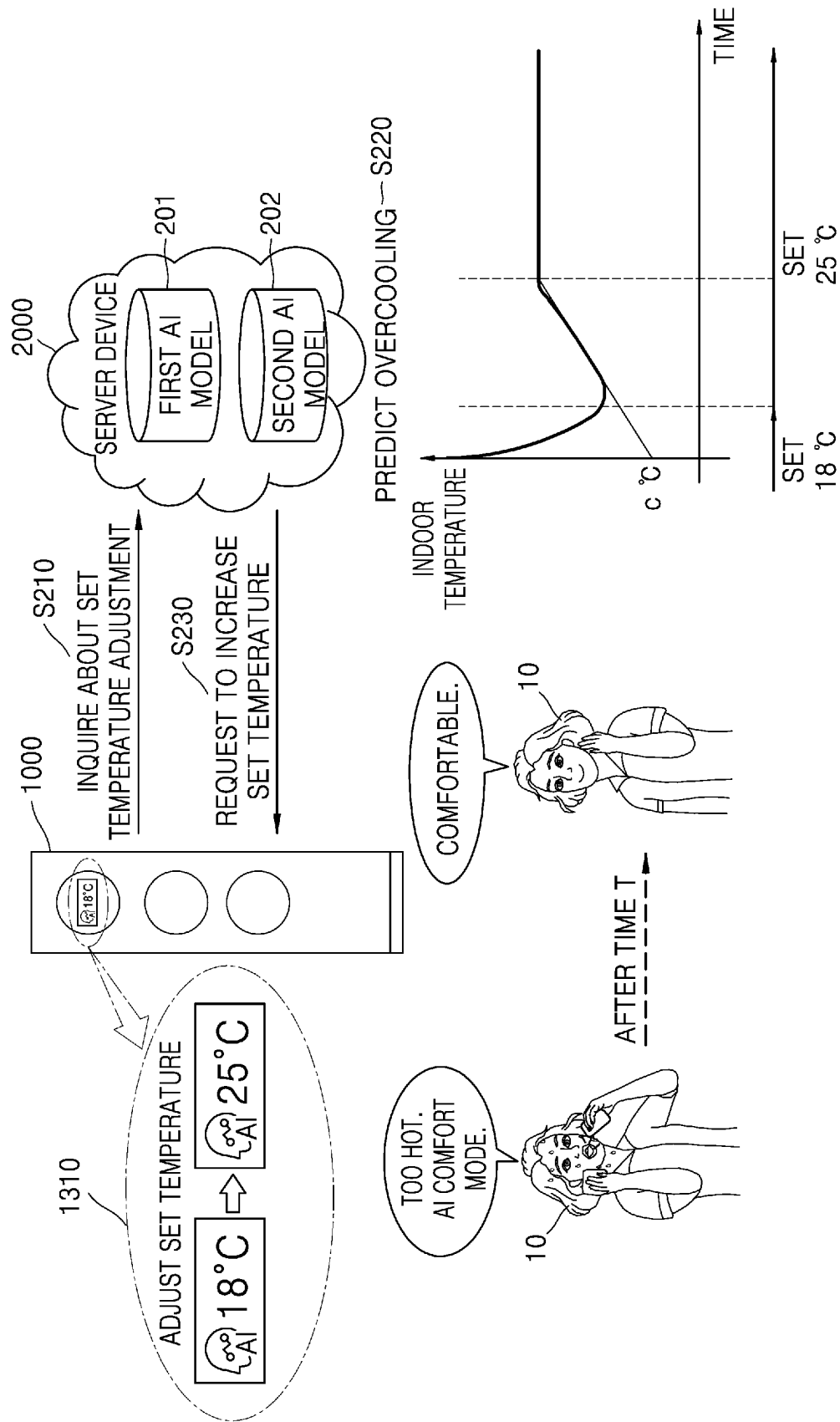


FIG. 3

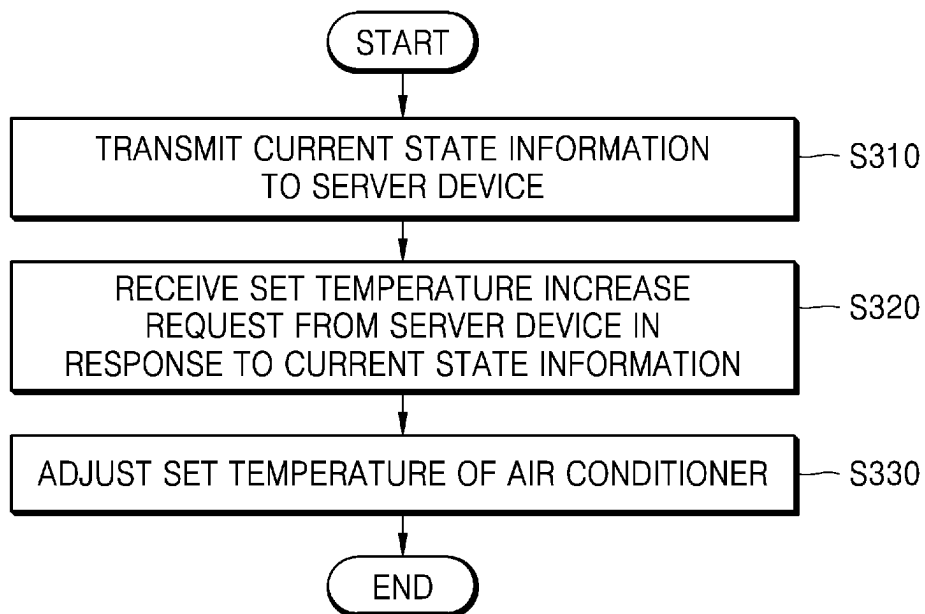


FIG. 4

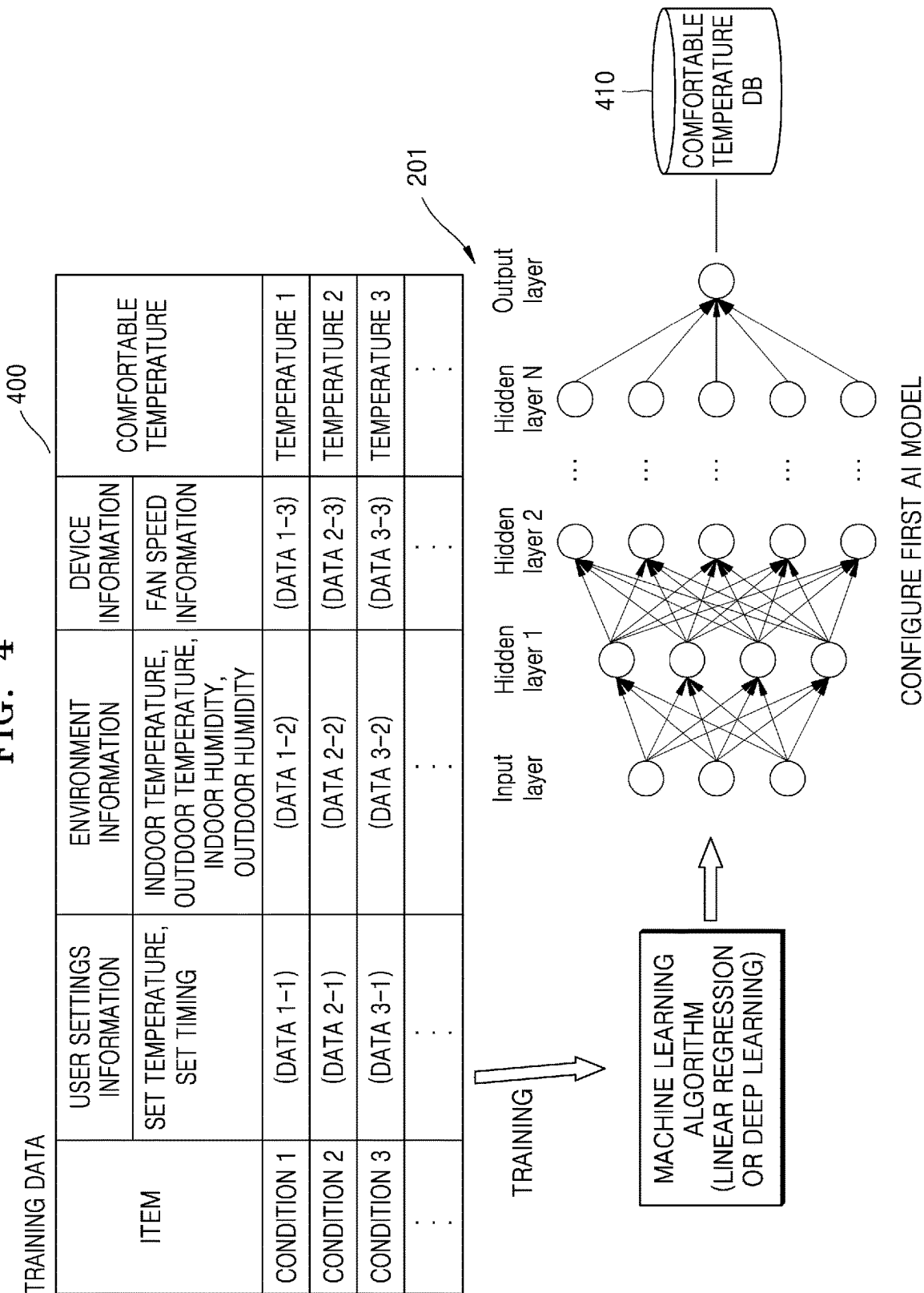


FIG. 5

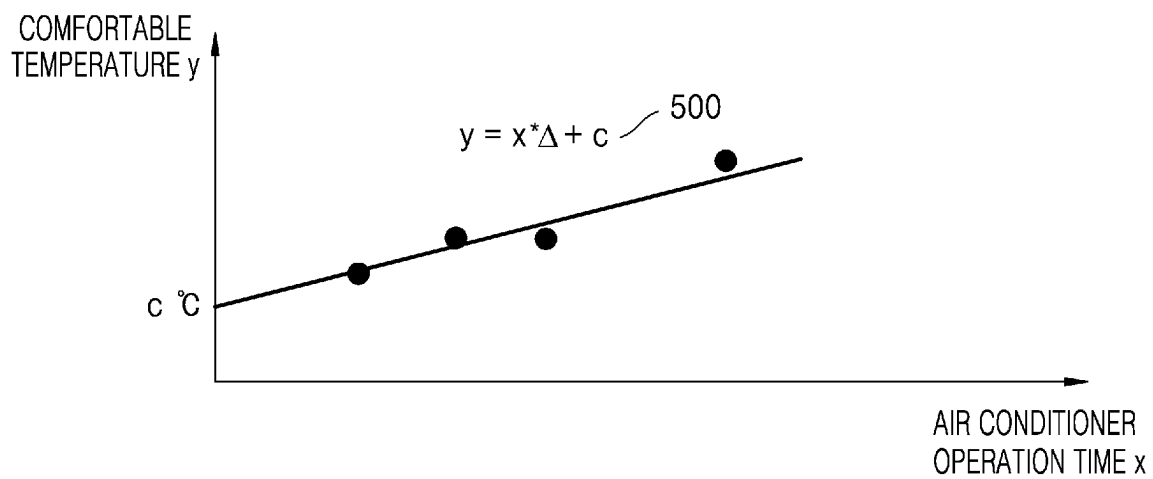


FIG. 6

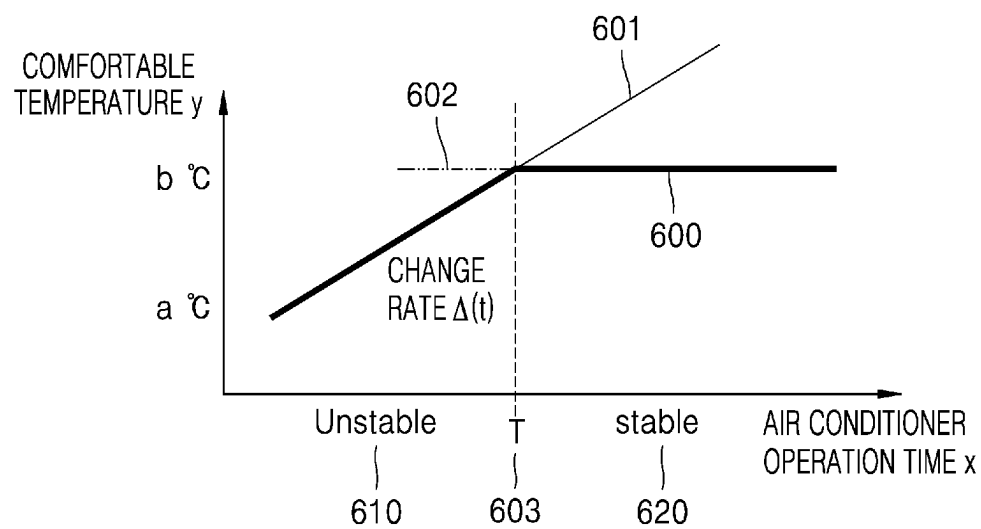


FIG. 7A

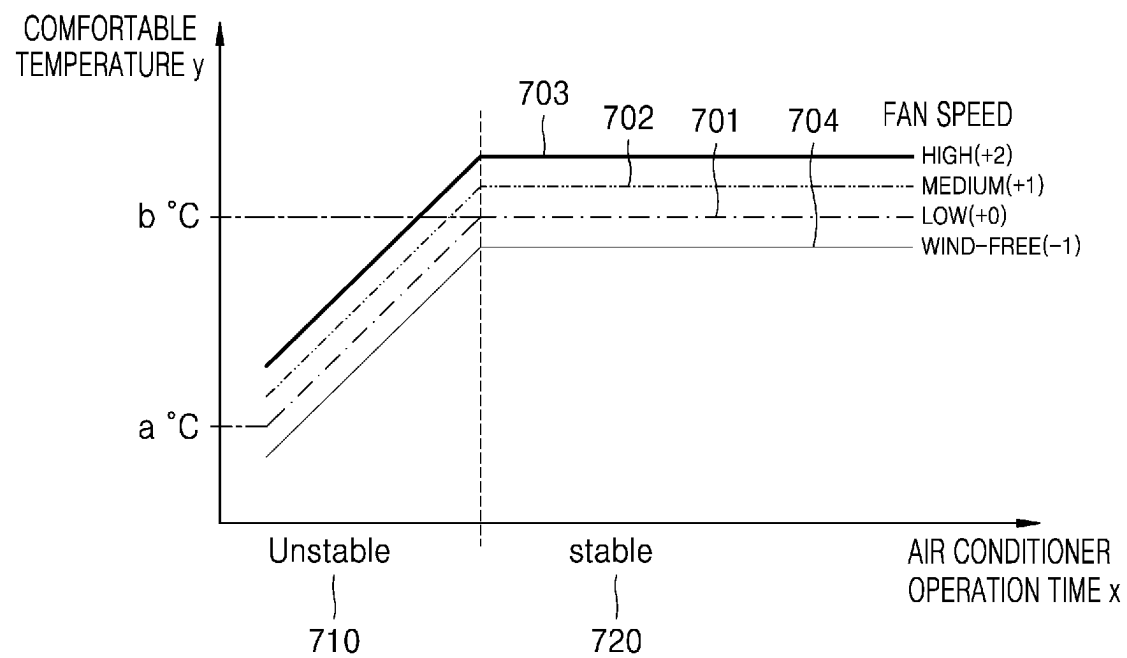


FIG. 7B

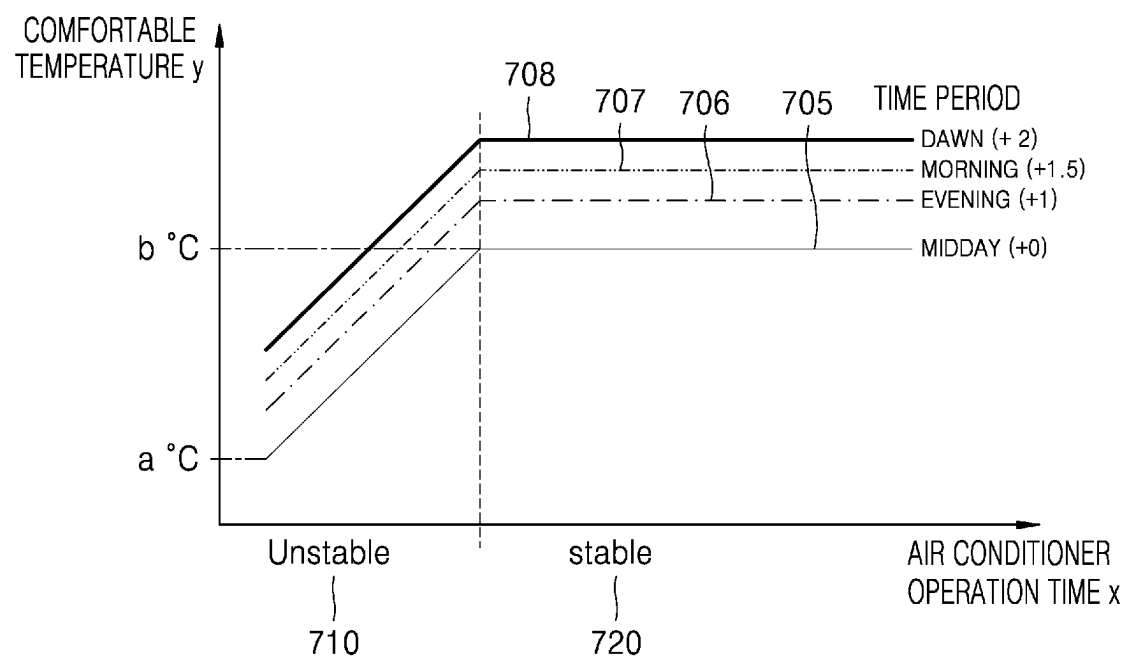


FIG. 8

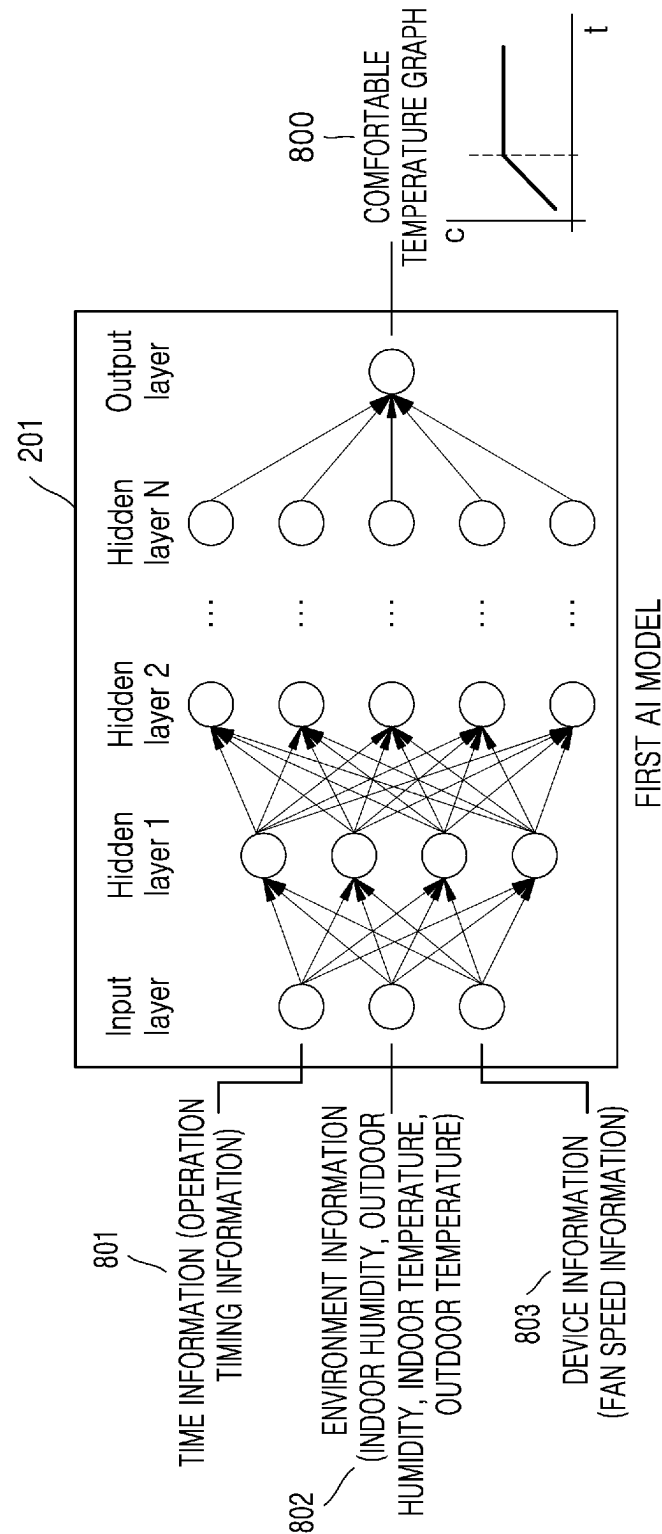


FIG. 9

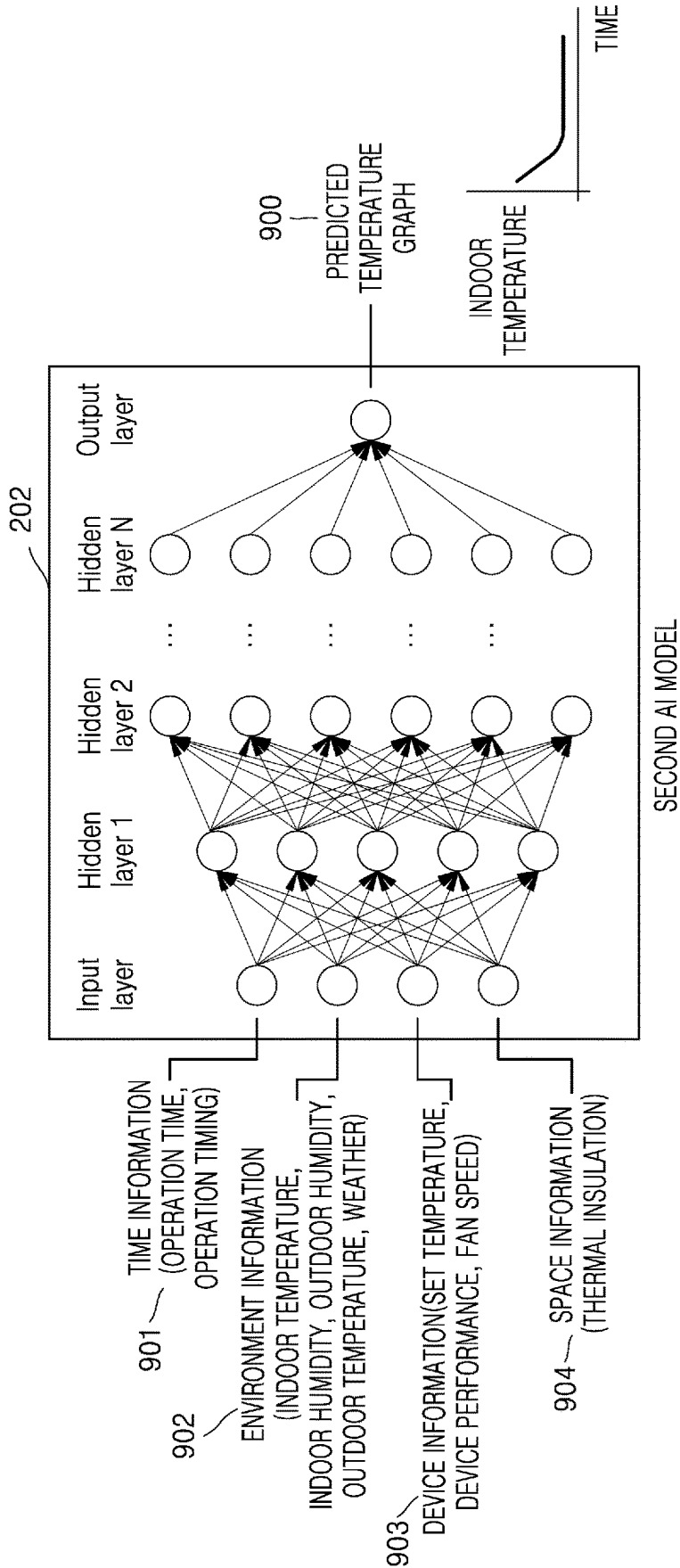


FIG. 10

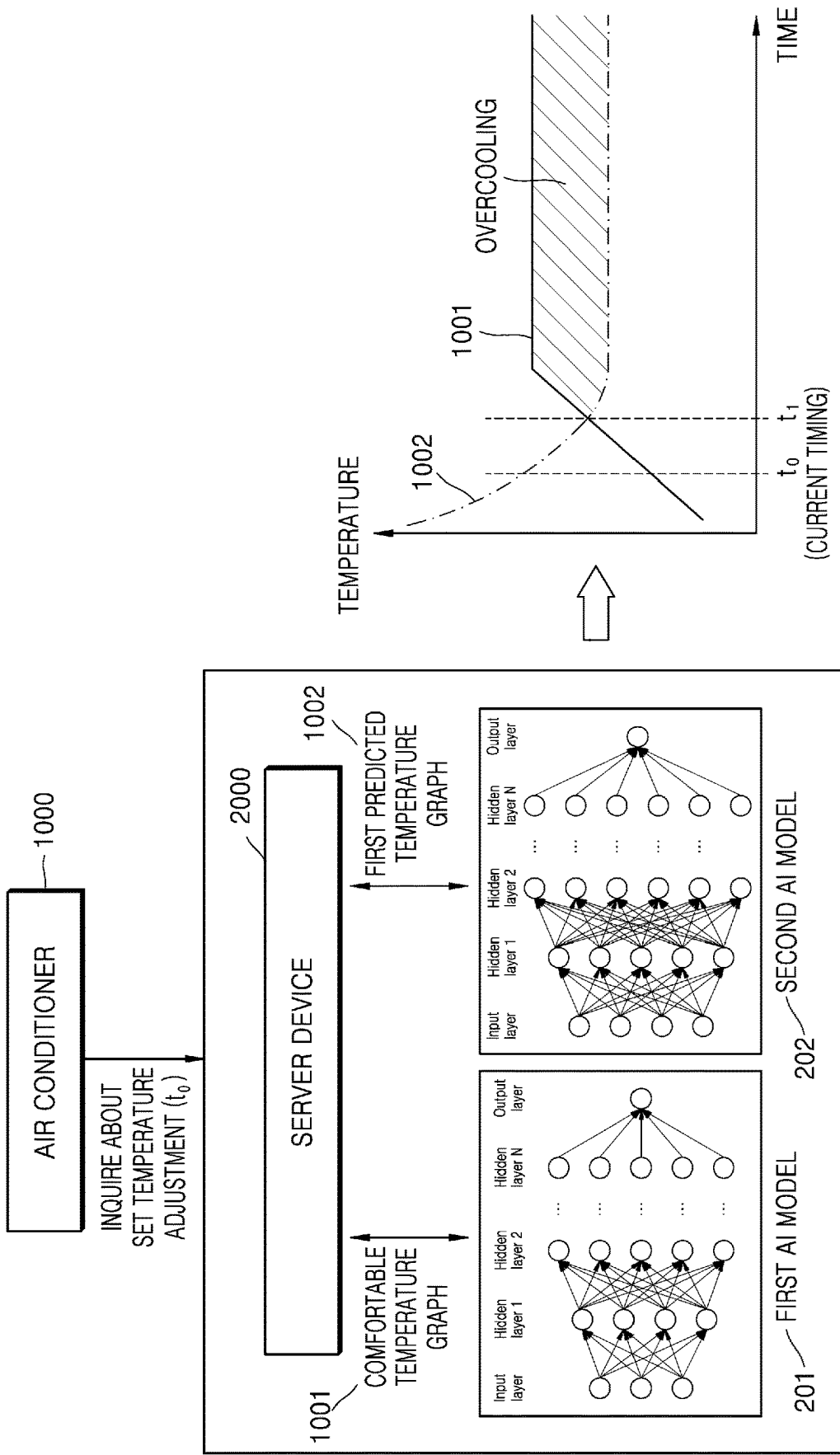


FIG. 11

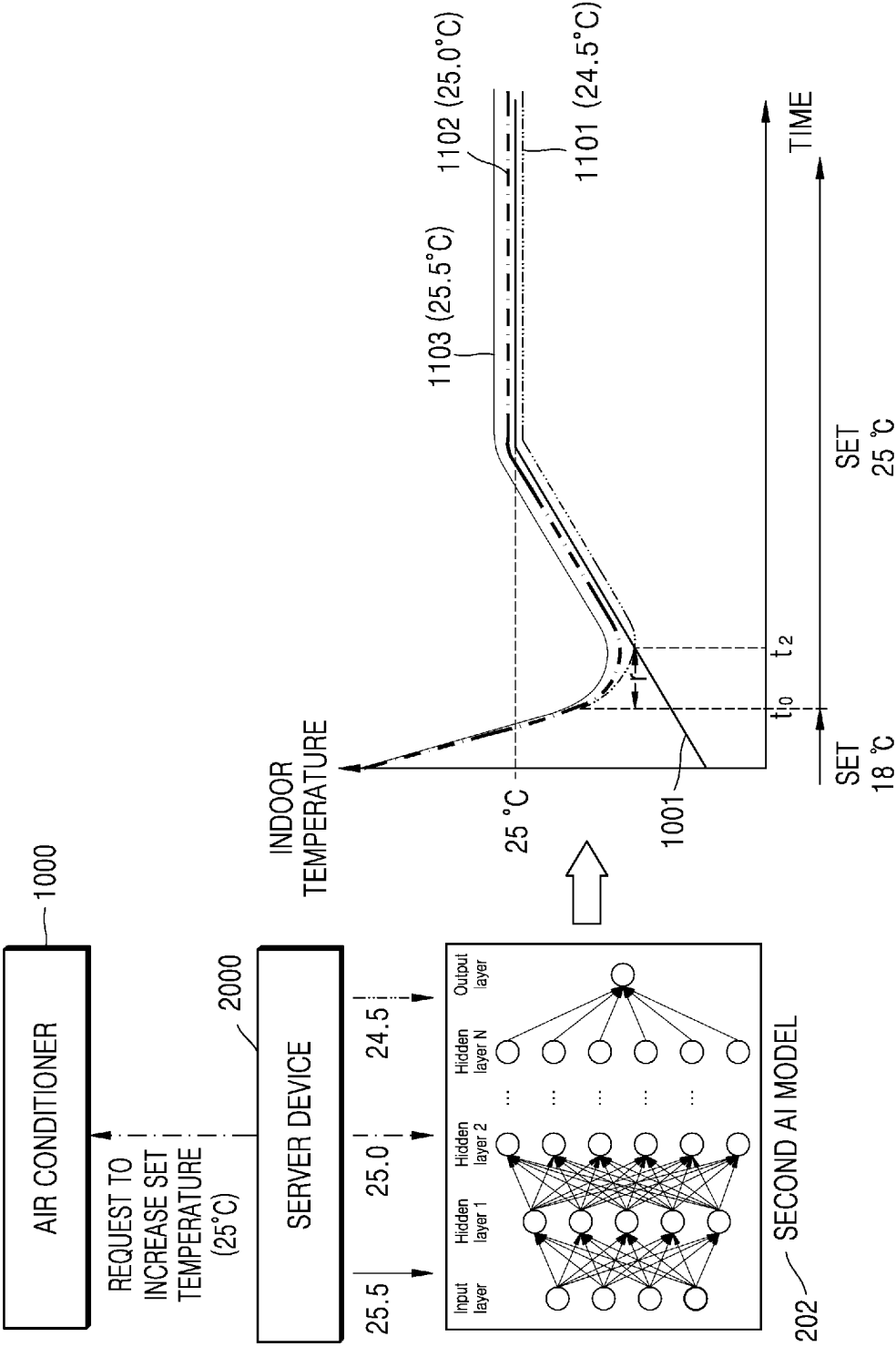


FIG. 12

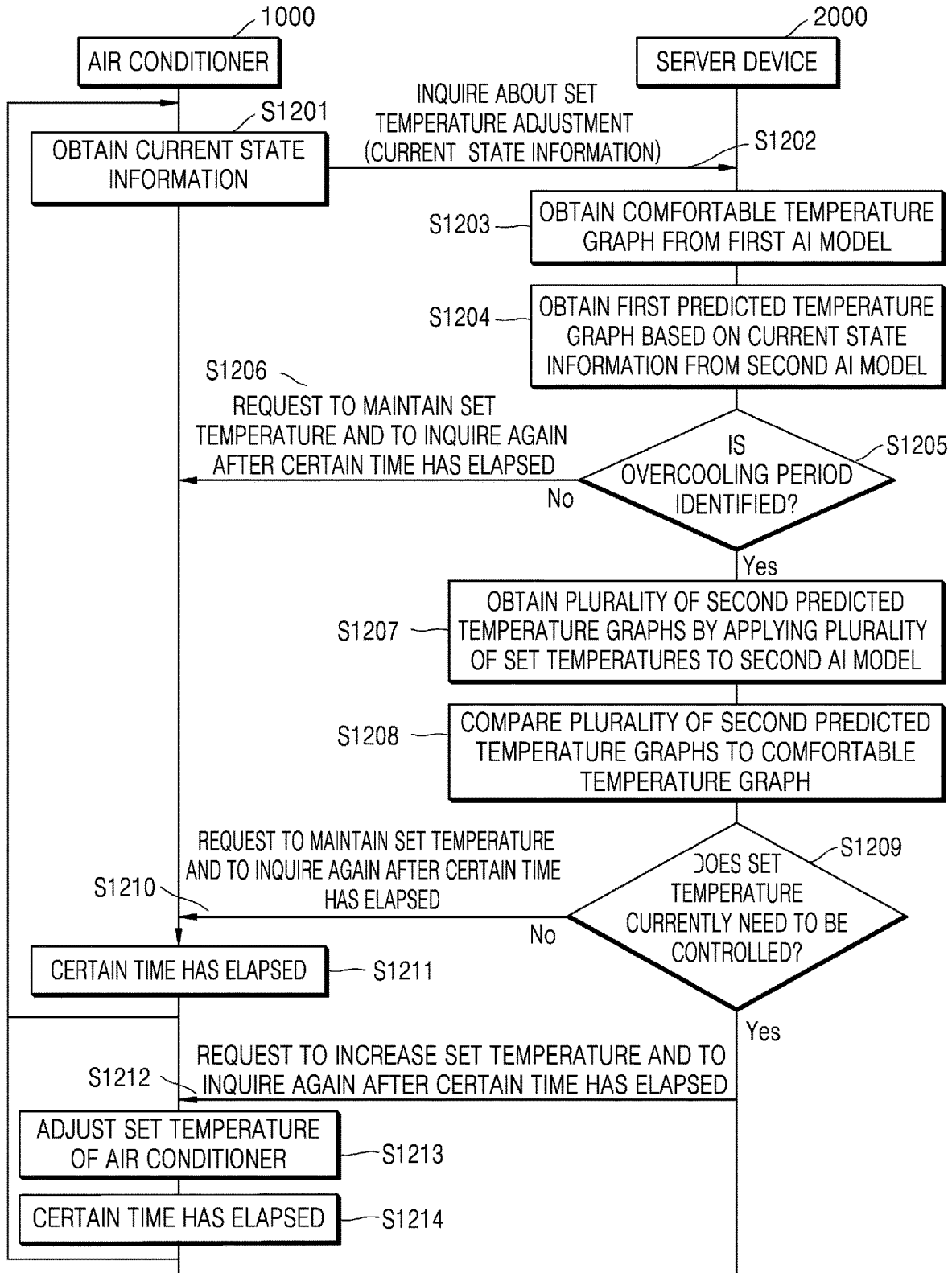


FIG. 13

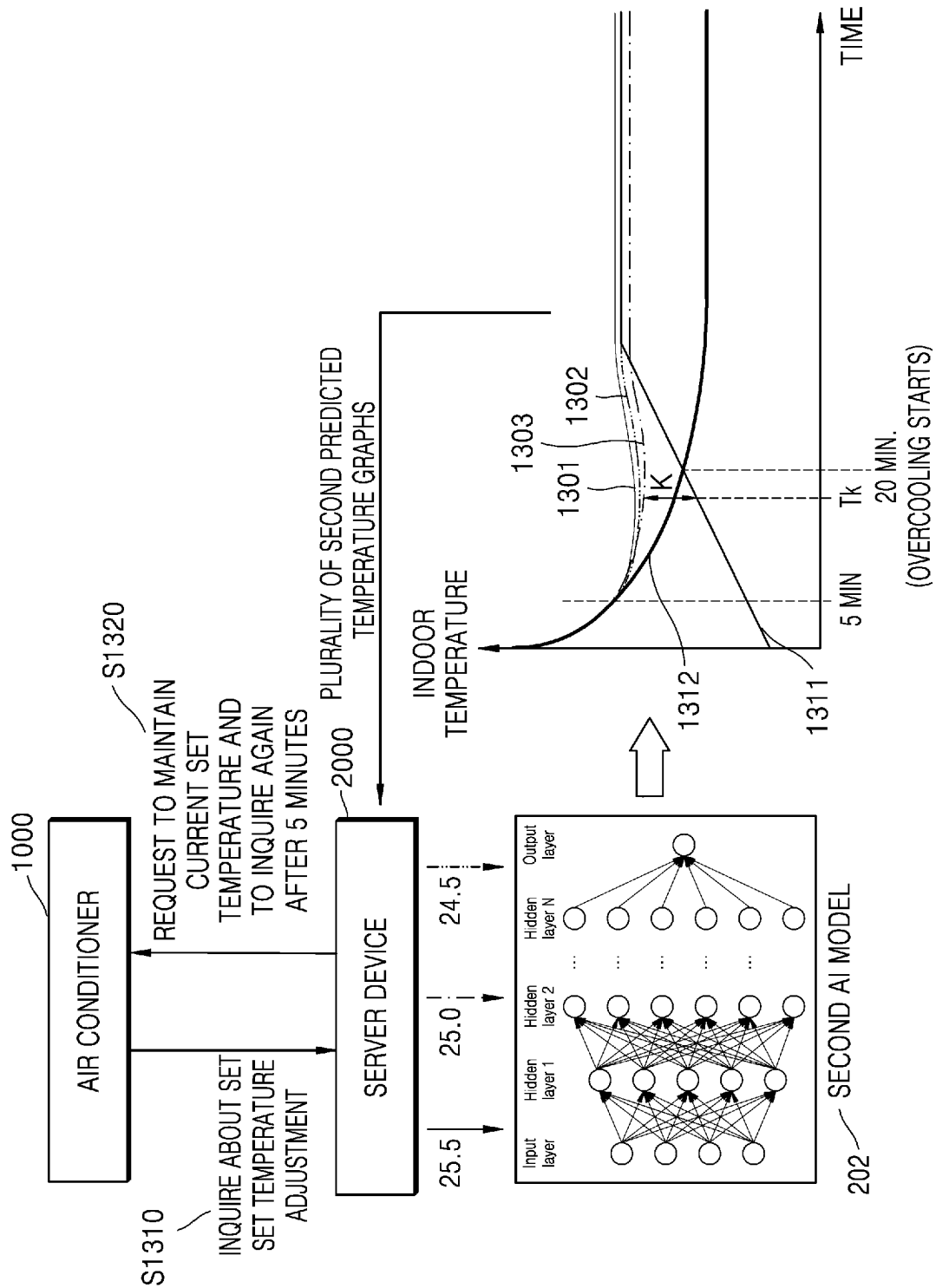


FIG. 14

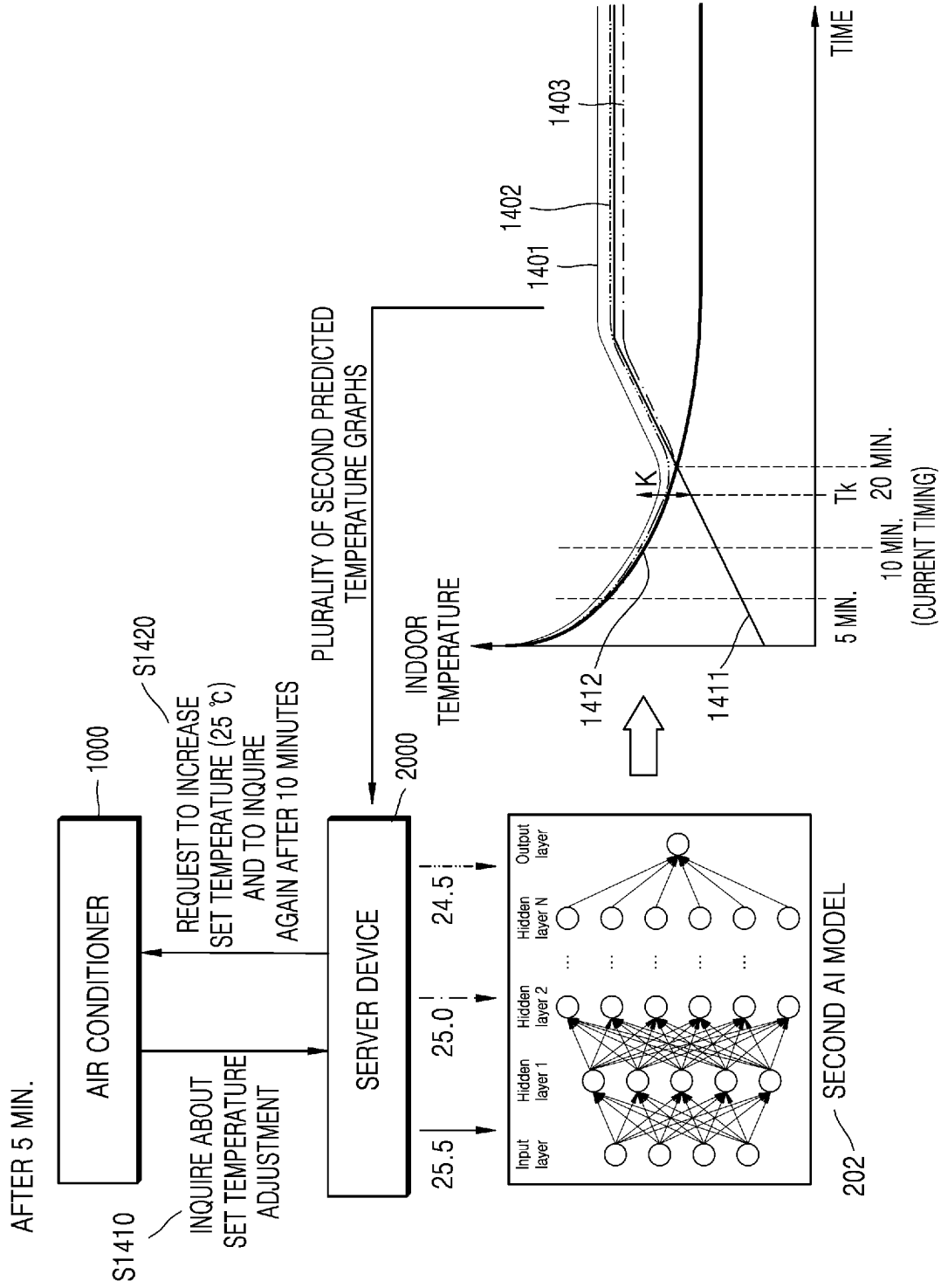


FIG. 15A

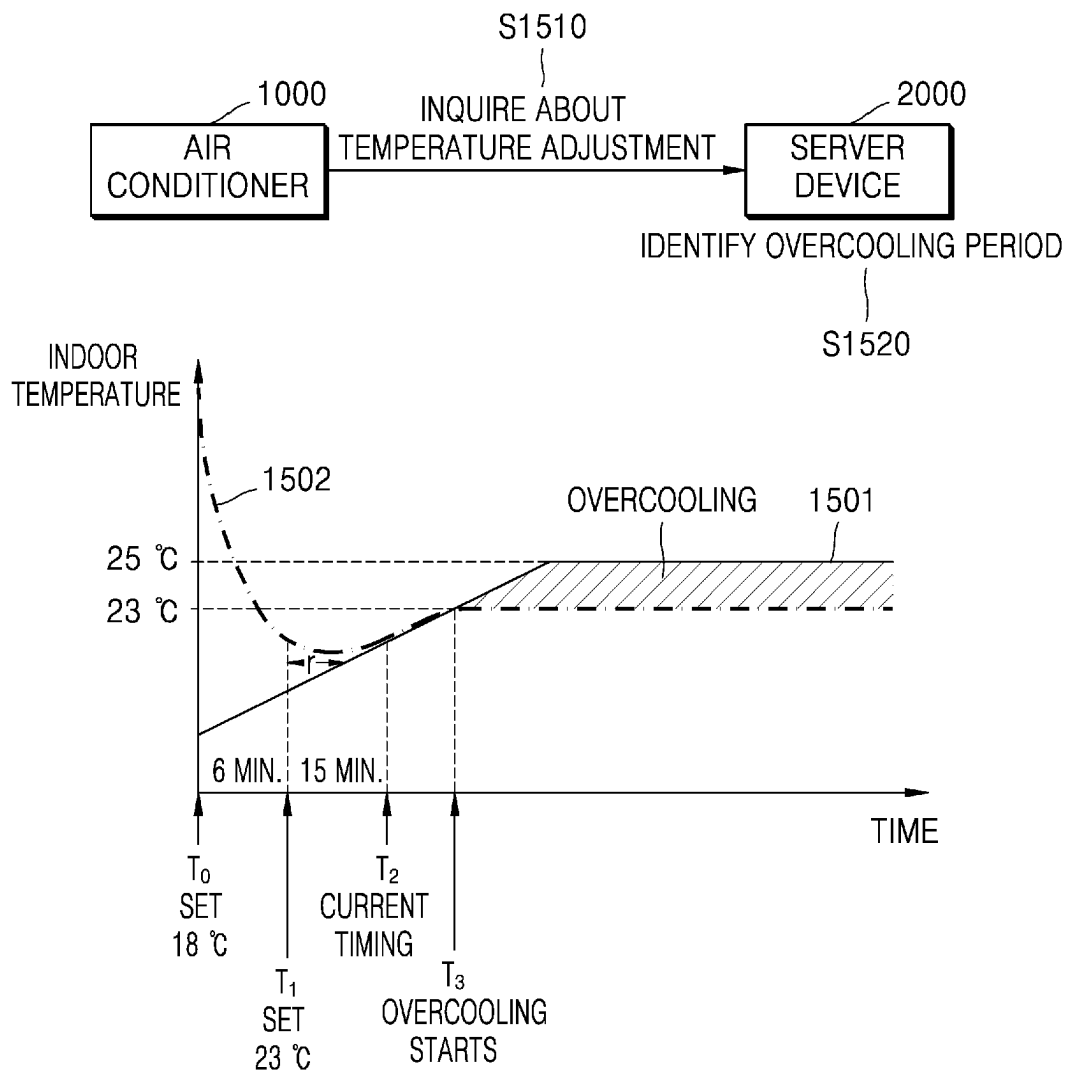


FIG. 15B

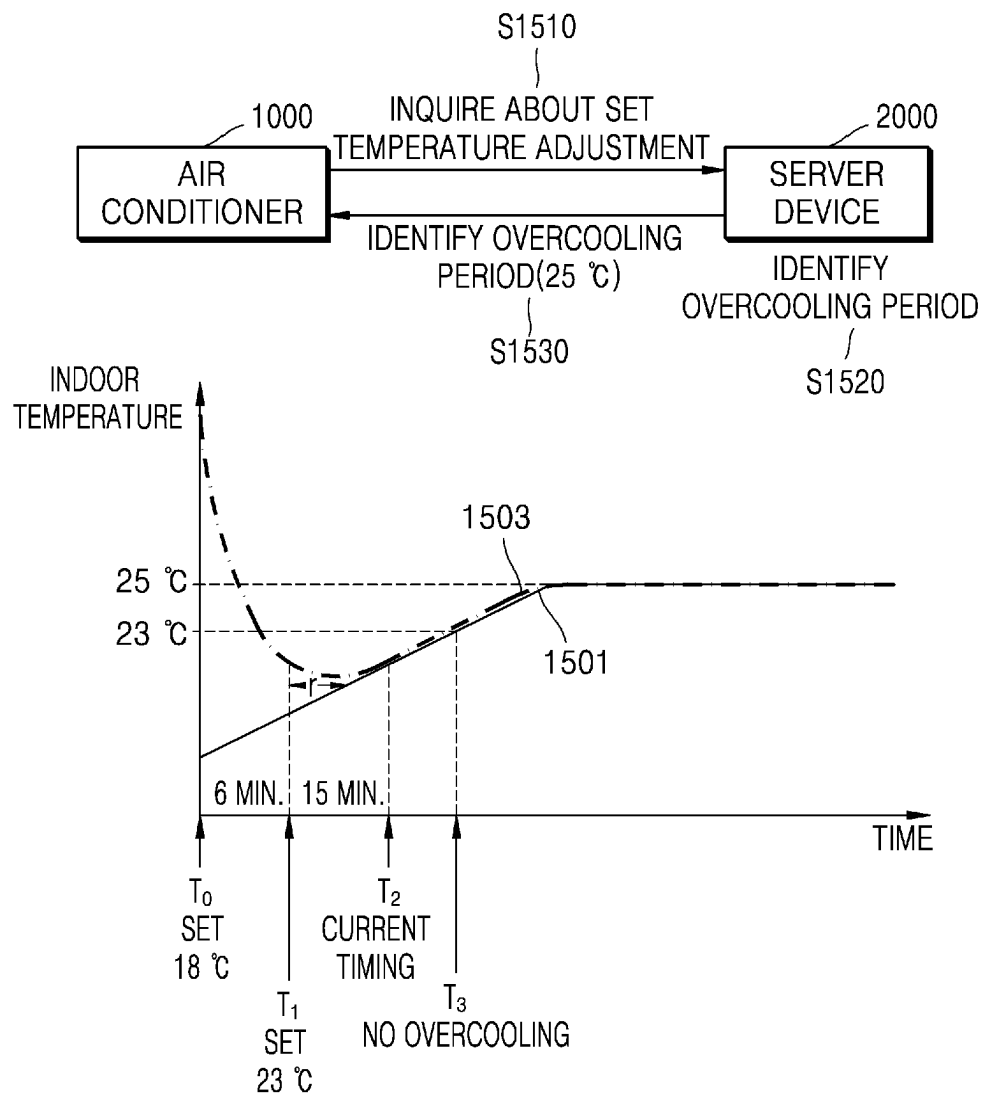


FIG. 16

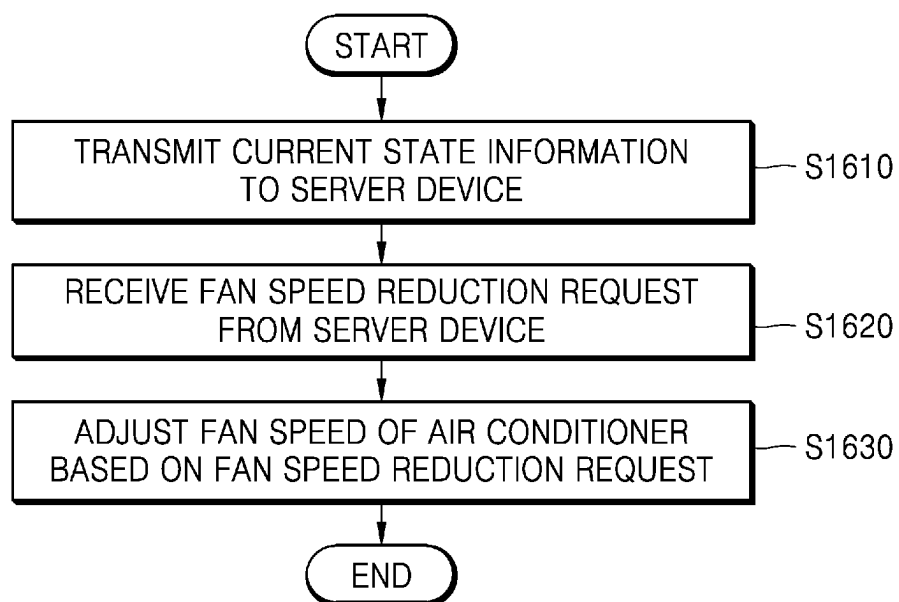


FIG. 17

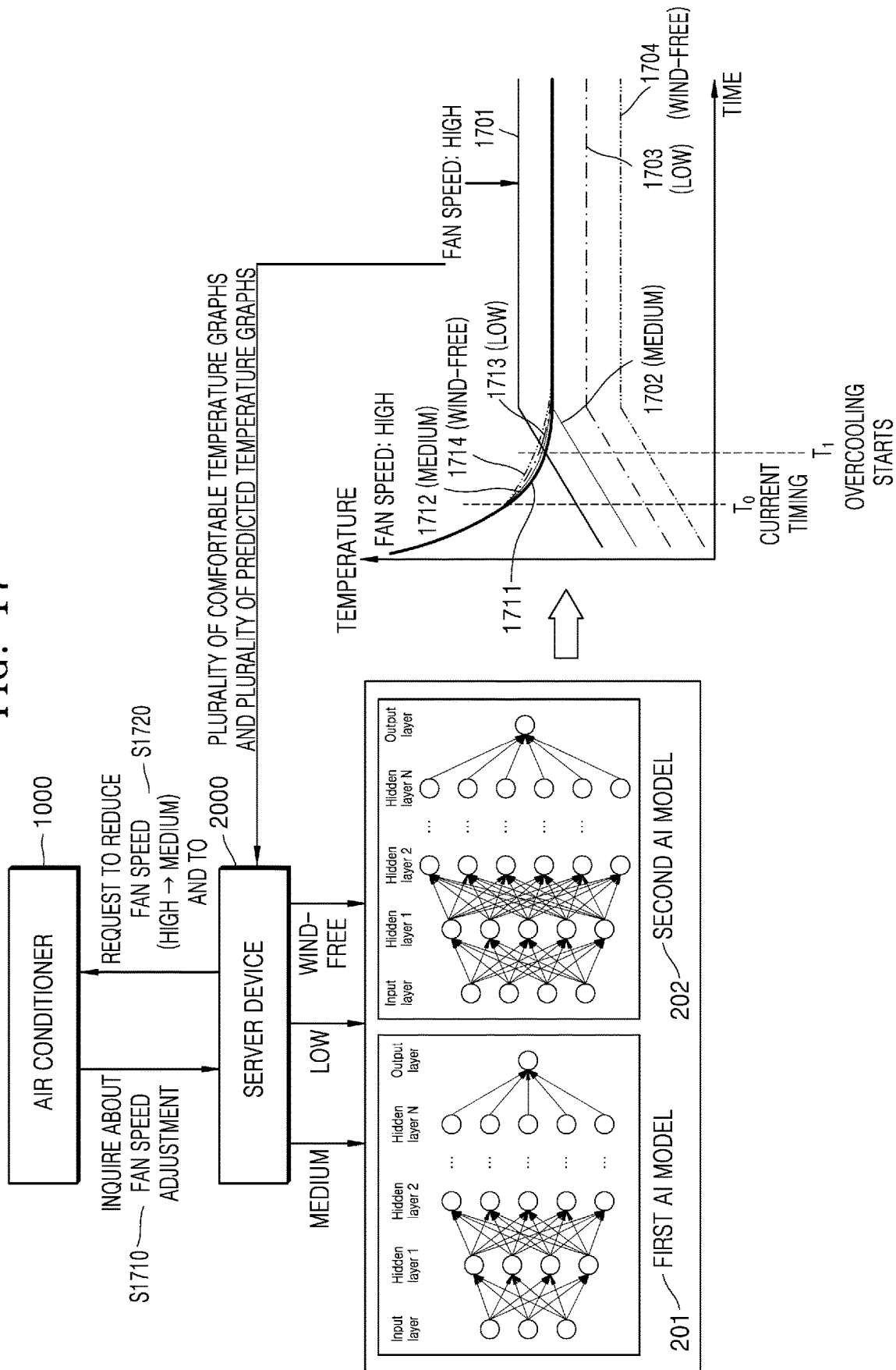


FIG. 18

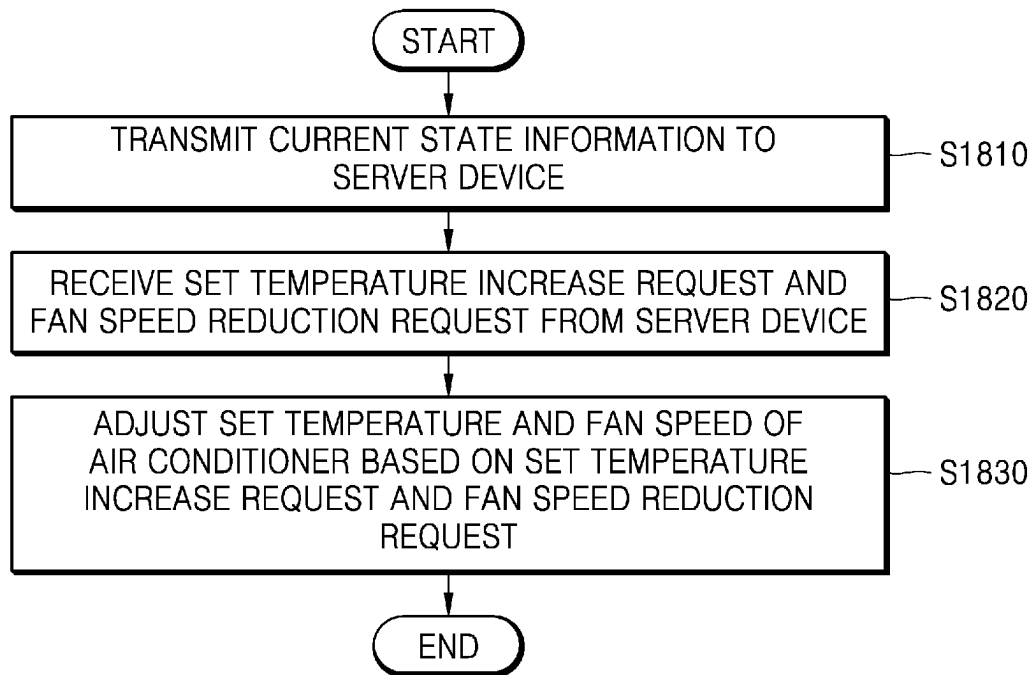


FIG. 19A

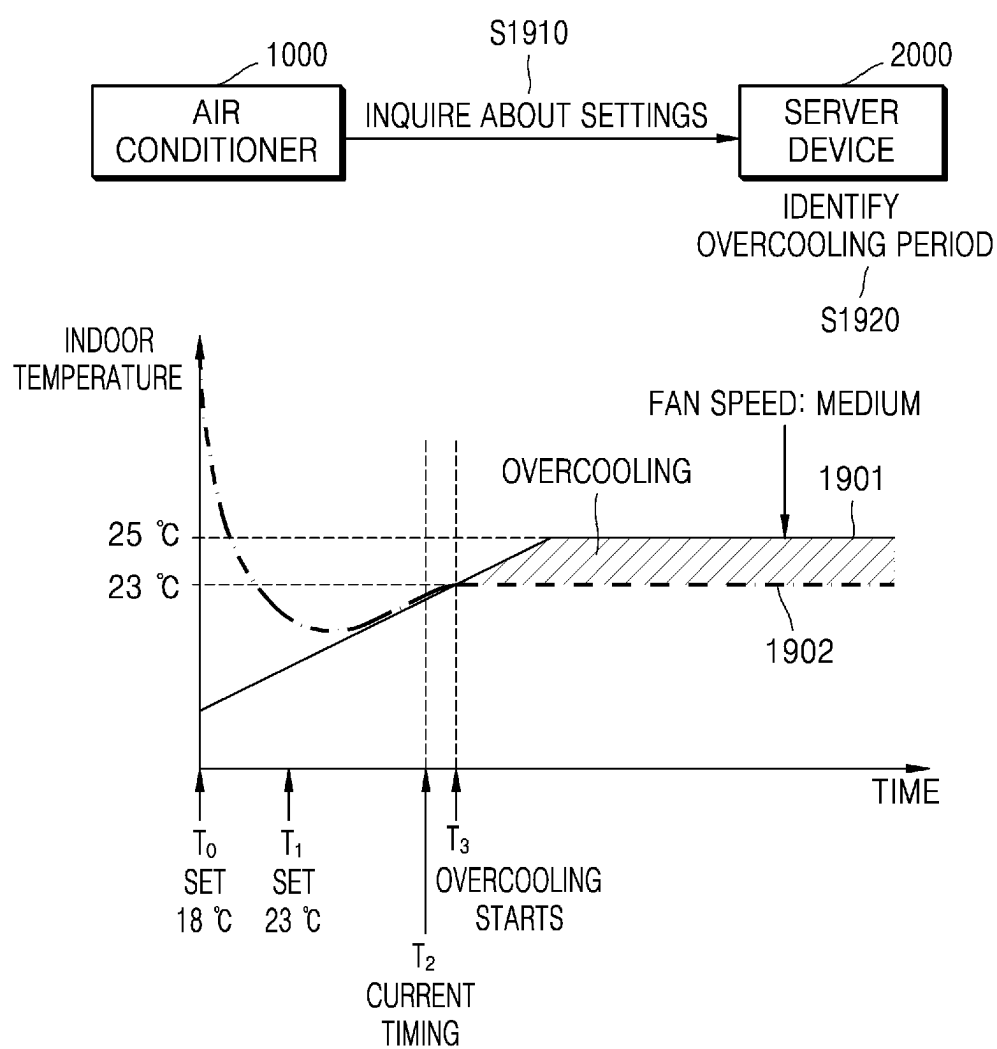


FIG. 19B

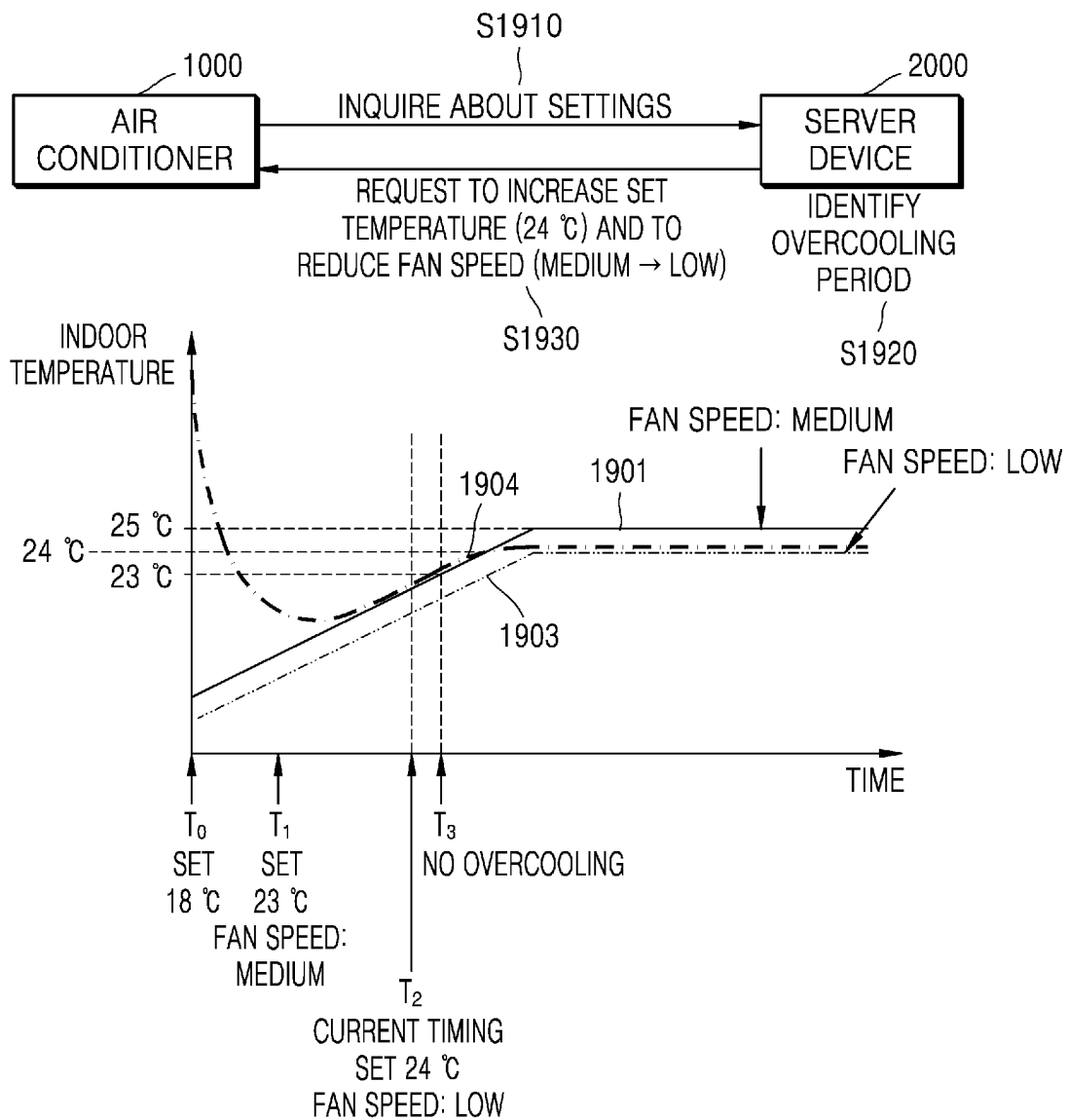


FIG. 20

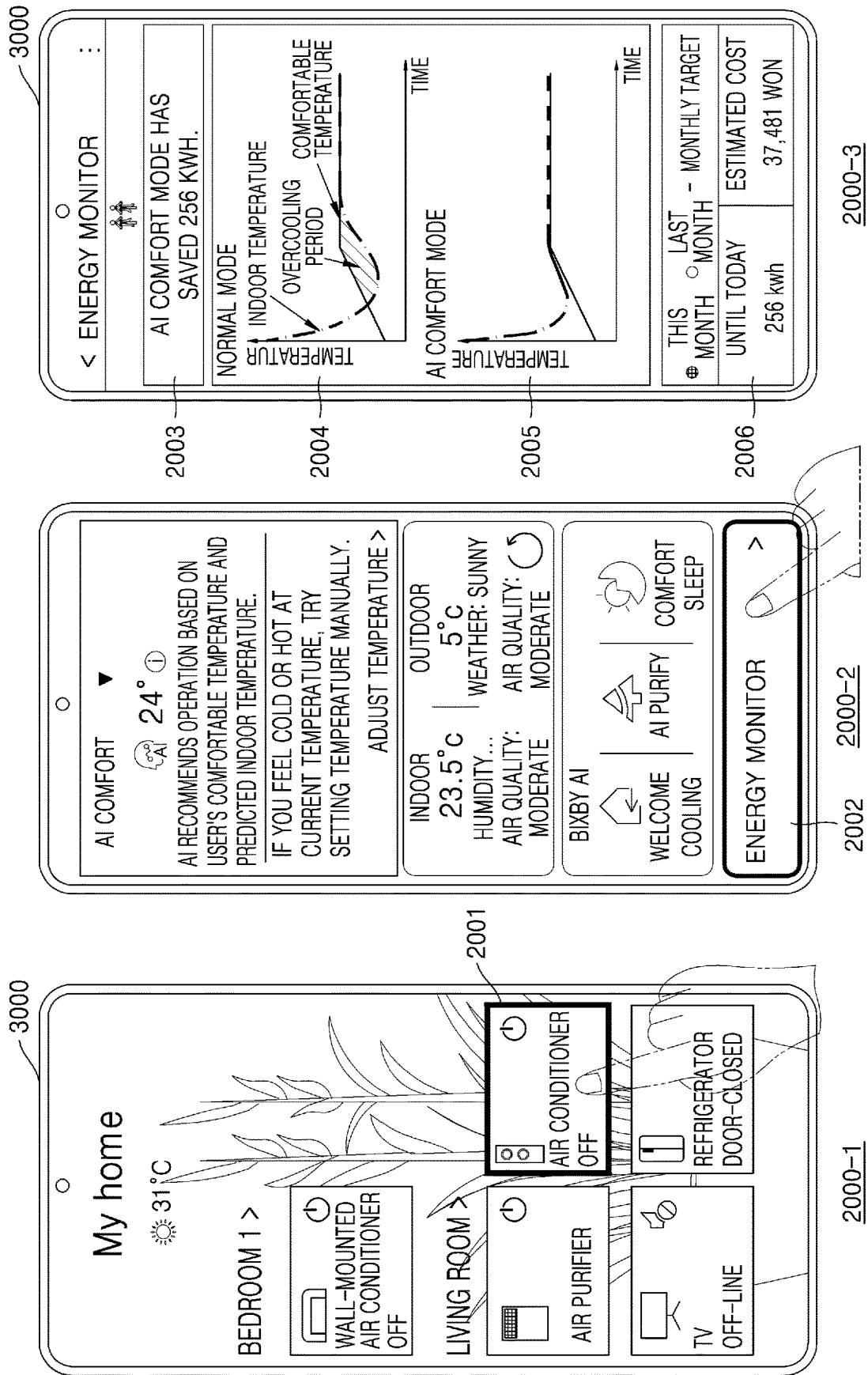


FIG. 21

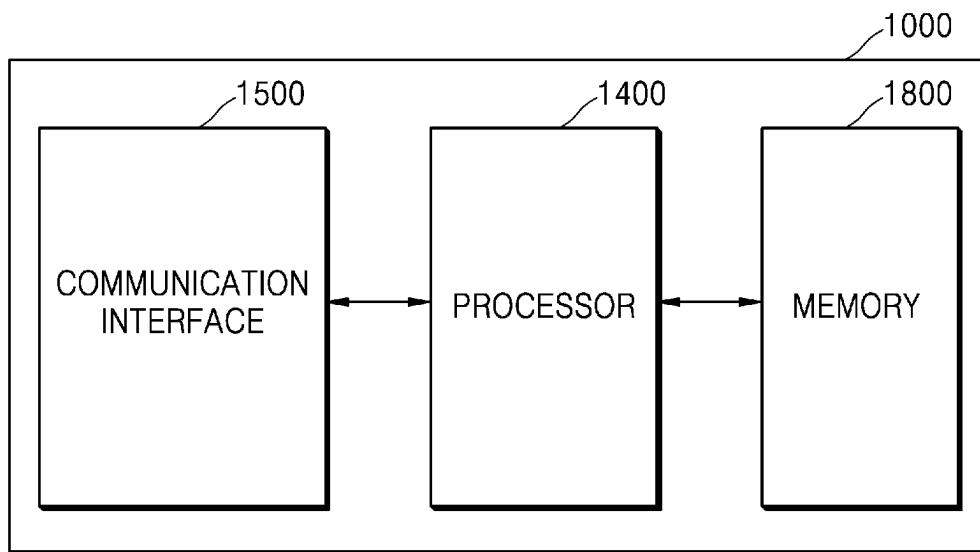


FIG. 22

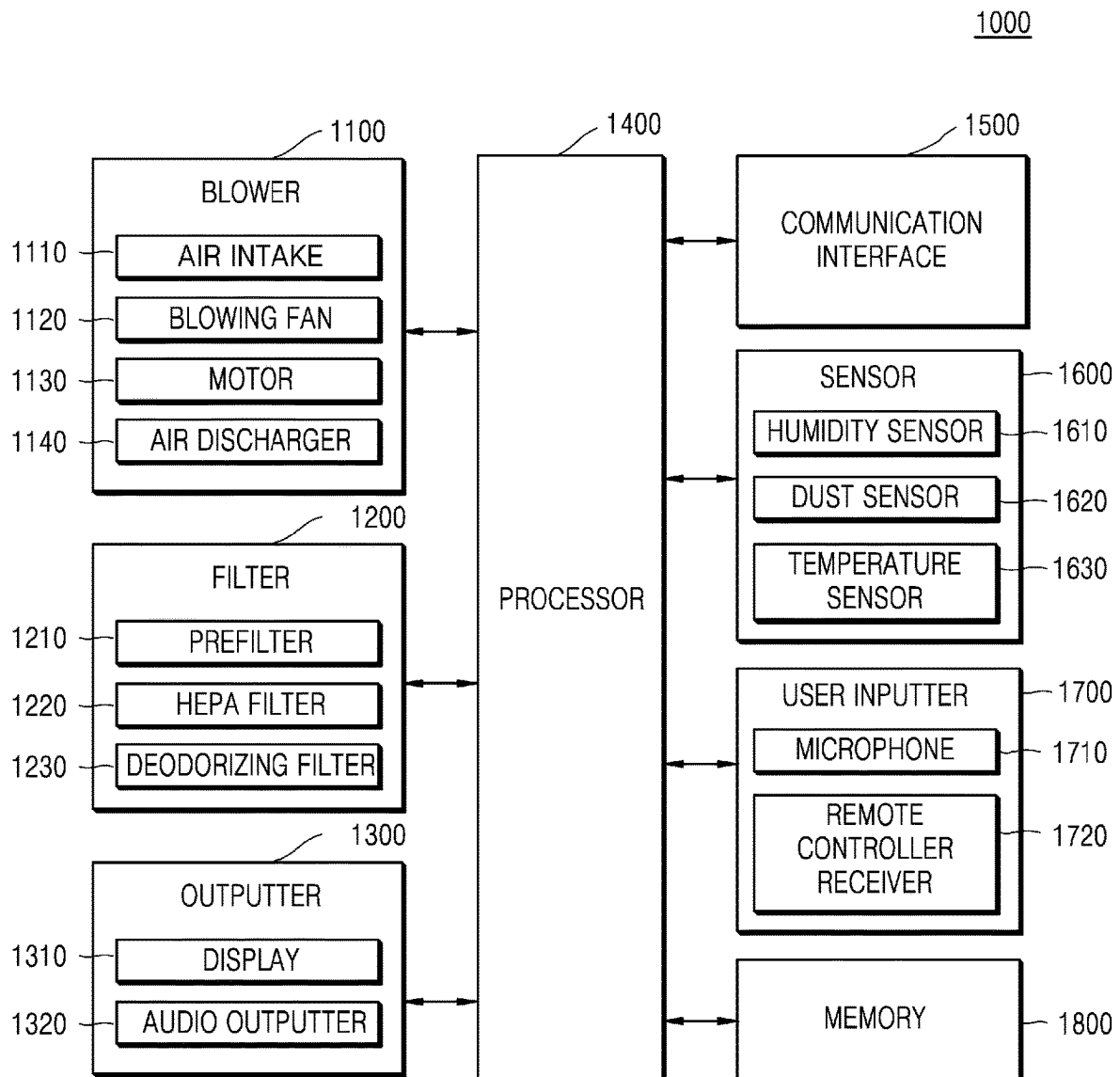
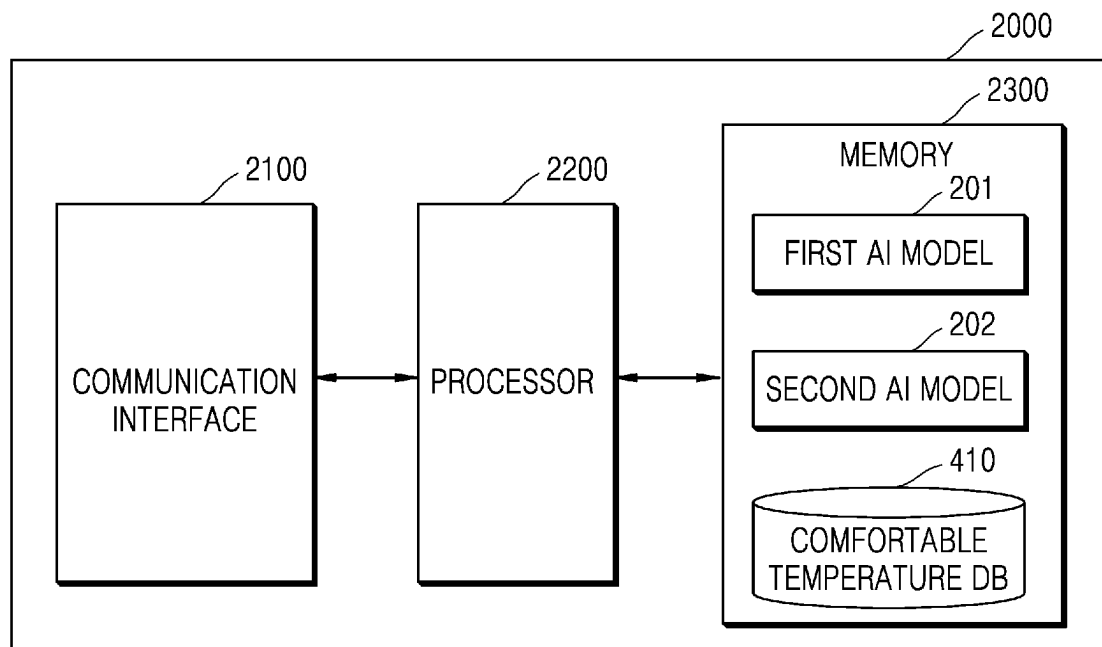


FIG. 23



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AIR CONDITIONER, AIR CONDITIONER CONTROL METHOD, AND AIR CONDITIONER CONTROL SYSTEM

CROSS-REFERENCE TO RELATED APPLICATION

This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Applications No. 10-2021-0071088, filed on Jun. 1, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety

BACKGROUND

1. Field

The disclosure relates to an air conditioner, an operating method of the air conditioner, and a control system including the air conditioner and a server device. More particularly, the disclosure relates to a control system for optimizing a set temperature or a fan speed of an air conditioner by using artificial intelligence (AI) models.

2. Description of the Related Art

Artificial intelligence (AI) systems are computer systems capable of implementing human-level intelligence, and refer to systems by which a machine autonomously learns, makes decisions, and becomes smarter, unlike existing rule-based smart systems. Because the AI systems may increase a recognition rate and more accurately understand user preferences in proportion to the number of iterations, the existing rule-based smart systems are being gradually replaced by deep-learning-based AI systems.

AI technology includes machine learning (e.g., deep learning), and element technologies using machine learning.

Machine learning is an algorithm technology for autonomously classifying/learning features of input data, and the element technologies are technologies for mimicking functions of the human brain, e.g., recognition and decision making, by using a machine learning algorithm such as deep learning, and include linguistic understanding, visual understanding, reasoning/prediction, knowledge representation, operation control, etc.

Various fields using AI technology are as described below. Linguistic understanding is a technology for recognizing and applying/processing human languages/characters, and includes natural language processing, machine translation, dialog systems, question answering, speech recognition/synthesis, etc. Visual understanding is a technology for recognizing and processing objects like human vision, and includes object detection, object tracking, image search, human detection, scene understanding, spatial understanding, image enhancement, etc. Reasoning/prediction is a technology for logically performing reasoning and prediction based on information, and includes knowledge/probability-based reasoning, optimized prediction, preference-based planning, recommendation, etc. Knowledge representation is a technology for automating human experience information into knowledge data, and includes knowledge construction (e.g., data generation/classification), knowledge management (data utilization), etc. Operation control is a technology for controlling autonomous driving of vehicles or motion of robots, and includes motion control (e.g., navigation, collision avoidance, and driving control), manipulation control (e.g., action control), etc.

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Currently, to provide a comfortable environment to a user, many attempts are being made to apply AI technology to home appliances such as robot vacuums, air conditioners, air purifiers, washers, and dryers.

SUMMARY

According to various embodiments of the disclosure, a method and system for optimizing a set temperature or a fan speed of an air conditioner may be provided. More particularly, according to various embodiments of the disclosure, a method and system for controlling a set temperature or a fan speed of an air conditioner by using artificial intelligence (AI) models to prevent an overcooling period in which an indoor temperature is reduced below a comfortable temperature may be provided.

According to an embodiment of the disclosure, an air conditioner includes a communication interface to communicate with a server device, a memory to store one or more instructions, and at least one processor. The at least one processor of the air conditioner is configured to execute the one or more instructions to transmit, to the server device through the communication interface, current state information including one or more of operation time information of the air conditioner, set temperature information of the air conditioner, and current indoor temperature information, receive a set temperature increase request corresponding to the transmitted current state information from the server device through the communication interface, and adjust a set temperature of the air conditioner based on the received set temperature increase request. A comfortable temperature graph is obtained from a first Artificial Intelligence (AI) model and a first predicted temperature graph is obtained from a second AI model based on the transmitted current state information, and the set temperature increase request is received when an overcooling period, in which the obtained first predicted temperature graph is reduced below the obtained comfortable temperature graph, is identified.

The comfortable temperature graph comprises an unstable period in which a comfortable temperature changes based on an operation time of the air conditioner, and a stable period in which the comfortable temperature is constantly maintained.

The comfortable temperature graph is obtained by applying one or more of operation timing information of the air conditioner, fan speed information of the air conditioner, indoor humidity information, outdoor humidity information, indoor temperature information, and outdoor temperature information to the first AI model.

The set temperature increase request comprises information instructing to change the set temperature of the air conditioner to a first temperature, and the first temperature is determined from among one or more set temperatures, based on a result of comparing the comfortable temperature graph to one or more second predicted temperature graphs obtained by inputting the one or more set temperatures to the second AI model.

A second predicted temperature graph obtained by applying the first temperature to the second AI model is not reduced below the comfortable temperature graph and is closest to the comfortable temperature graph.

The processor is further configured to execute the one or more instructions to receive a fan speed reduction request from the server device corresponding to the transmitted current state information, and adjust a fan speed of the air conditioner based on the fan speed reduction request.

The first predicted temperature graph is obtained by further applying one or more of operation timing information of the air conditioner, indoor humidity information, outdoor temperature information, outdoor humidity information, weather information, performance information of the air conditioner, and installation space information of the air conditioner to the second AI model.

According to an embodiment of the disclosure, an operating method of an air conditioner includes transmitting, to a server device, current state information including one or more of operation time information of the air conditioner, set temperature information of the air conditioner, and current indoor temperature information, receiving a set temperature increase request corresponding to the transmitted current state information from the server device, and adjusting a set temperature of the air conditioner based on the set temperature increase request, a comfortable temperature graph is obtained from a first Artificial Intelligence (AI) model and a first predicted temperature graph is obtained from a second AI model based on the transmitted current state information, and the set temperature increase request is received when an overcooling period, in which the obtained first predicted temperature graph is reduced below the obtained comfortable temperature graph, is identified.

The comfortable temperature graph may include an unstable period in which a comfortable temperature changes based on an operation time of the air conditioner, and a stable period in which the comfortable temperature is constantly maintained, and be obtained by applying one or more of operation timing information of the air conditioner, fan speed information of the air conditioner, indoor humidity information, outdoor humidity information, indoor temperature information, and outdoor temperature information to the first AI model.

The set temperature increase request may include information instructing to change the set temperature of the air conditioner to a first temperature, and the server device may be further configured to determine the first temperature from among one or more set temperatures, based on a result of comparing the comfortable temperature graph to one or more second predicted temperature graphs obtained by inputting the one or more set temperatures to the second AI model.

A second predicted temperature graph obtained by applying the first temperature to the second AI model may not be reduced below the comfortable temperature graph and be closest to the comfortable temperature graph.

The air conditioner may be further configured to receive a fan speed reduction request from the server device corresponding to the transmitted current state information, and adjust a fan speed of the air conditioner based on the fan speed reduction request.

The server device may be further configured to obtain the first predicted temperature graph from the second AI model by further applying one or more of operation timing information of the air conditioner, indoor humidity information, outdoor temperature information, outdoor humidity information, weather information, performance information of the air conditioner, or installation space information of the air conditioner to the second AI model.

A control system comprises an air conditioner configured to transmit, to a server device, current state information comprising one or more of operation time information of the air conditioner, set temperature information of the air conditioner, and current indoor temperature information, and adjust a set temperature of the air conditioner based on a set temperature increase request, corresponding to the transmit-

ted current state information, received from the server device, and the server device configured to transmit the set temperature increase request to the air conditioner when an overcooling period, is identified.

A comfortable temperature graph is obtained from a first Artificial Intelligence (AI) model and a first predicted temperature graph is obtained from a second AI model based on the transmitted current state information, and the overcooling period is a period when the obtained first predicted temperature is reduced below the obtained comfortable temperature graph.

The comfortable temperature graph comprises an unstable period in which a comfortable temperature changes based on an operation time of the air conditioner, and a stable period in which the comfortable temperature is constantly maintained, and is obtained by applying one or more of operation timing information of the air conditioner, fan speed information of the air conditioner, indoor humidity information, outdoor humidity information, indoor temperature information, and outdoor temperature information to the first AI model.

The set temperature increase request comprises information instructing to change the set temperature of the air conditioner to a first temperature, and the server device is further configured to determine the first temperature from among one or more set temperatures, based on a result of comparing the comfortable temperature graph to one or more second predicted temperature graphs obtained by inputting the one or more set temperatures to the second AI model.

A second predicted temperature graph obtained by applying the first temperature to the second AI model is not reduced below the comfortable temperature graph and is closest to the comfortable temperature graph.

The air conditioner is further configured to receive a fan speed reduction request from the server device corresponding to the transmitted current state information, and adjust a fan speed of the air conditioner based on the fan speed reduction request.

The server device is further configured to obtain the first predicted temperature graph from the second AI model by further applying one or more of operation timing information of the air conditioner, indoor humidity information, outdoor temperature information, outdoor humidity information, weather information, performance information of the air conditioner, and installation space information of the air conditioner to the second AI model.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a diagram for describing a general air conditioner control system.

FIG. 2 is a diagram for describing an air conditioner control system according to an embodiment of the disclosure.

FIG. 3 is a flowchart of an operating method of an air conditioner, according to an embodiment of the disclosure.

FIG. 4 is a diagram for describing an operation of training a first artificial intelligence (AI) model, according to an embodiment of the disclosure.

FIG. 5 is a diagram for describing an operation of modeling an unstable period of a comfortable temperature graph, according to an embodiment of the disclosure.

FIG. 6 is a diagram for describing an unstable period and a stable period of a comfortable temperature graph, according to an embodiment of the disclosure.

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FIG. 7A is a diagram for describing a plurality of comfortable temperature graphs obtained by applying a plurality of fan speed values to a first AI model, according to an embodiment of the disclosure.

FIG. 7B is a diagram for describing a plurality of comfortable temperature graphs obtained by applying a plurality of time periods to a first AI model, according to an embodiment of the disclosure.

FIG. 8 is a diagram for describing an operation of obtaining a comfortable temperature graph from a first AI model, according to an embodiment of the disclosure.

FIG. 9 is a diagram for describing an operation of obtaining a predicted temperature graph from a second AI model, according to an embodiment of the disclosure.

FIG. 10 is a diagram for describing an operation, performed by a server device, of identifying an overcooling period, according to an embodiment of the disclosure.

FIG. 11 is a diagram for describing an operation, performed by a server device, of determining a set temperature of an air conditioner by using a second AI model, according to an embodiment of the disclosure.

FIG. 12 is a flowchart of a method of adjusting a set temperature of an air conditioner, according to an embodiment of the disclosure.

FIG. 13 is a diagram for describing an operation, performed by a server device, of requesting an air conditioner to inquire again after a certain time has elapsed, according to an embodiment of the disclosure.

FIG. 14 is a diagram for describing an operation, performed by an air conditioner, of inquiring of a server device about set temperature adjustment again after a certain time has elapsed, according to an embodiment of the disclosure.

FIGS. 15A and 15B are diagrams for describing an operation, performed by an air conditioner, of adjusting a set temperature a plurality of times, according to an embodiment of the disclosure.

FIG. 16 is a flowchart of a method, performed by an air conditioner, of adjusting a fan speed, according to an embodiment of the disclosure.

FIG. 17 is a diagram for describing an operation, performed by a server device, of determining a fan speed of an air conditioner by using a first AI model and a second AI model, according to an embodiment of the disclosure.

FIG. 18 is a flowchart of a method, performed by an air conditioner, of adjusting a set temperature and a fan speed, according to an embodiment of the disclosure.

FIGS. 19A and 19B are diagrams for describing an operation, performed by a server device, of determining a set temperature and a fan speed, according to an embodiment of the disclosure.

FIG. 20 is a diagram for describing an operation, performed by a server device, of providing energy saving information based on an AI comfort mode through a display device, according to an embodiment of the disclosure.

FIG. 21 is a block diagram for describing functions of an air conditioner, according to an embodiment of the disclosure.

FIG. 22 is a block diagram for describing functions of an air conditioner, according to an embodiment of the disclosure.

FIG. 23 is a block diagram for describing functions of a server device, according to an embodiment of the disclosure.

DETAILED DESCRIPTION

Throughout the disclosure, the expression “at least one of a, b or c” indicates only a, only b, only c, both a and b, both a and c, both b and c, all of a, b, and c, or variations thereof.

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Terminology used in this specification will now be briefly described before describing embodiments of the disclosure in detail.

Although the terms used herein are selected, as much as possible, from general terms that are widely used at present while taking into consideration the functions obtained in accordance with the disclosure, these terms may be replaced by other terms based on intentions of one of ordinary skill in the art, customs, emergence of new technologies, or the like. In a particular case, terms that are arbitrarily selected by the applicant may be used and, in that case, the meanings of these terms may be described in relevant parts of the disclosure. Therefore, it is noted that the terms used herein are construed based on practical meanings thereof and the whole content of this specification, rather than being simply construed based on the names of the terms.

It will be understood that the terms “comprises”, “comprising”, “includes” and/or “including”, when used herein, specify the presence of stated elements, but do not preclude the presence or addition of one or more other elements, unless otherwise indicated herein. As used herein, the term “unit” or “module” denotes an entity for performing at least one function or operation, and may be implemented as hardware, software, or a combination of hardware and software.

Hereinafter, the disclosure will be described in detail by explaining embodiments of the disclosure with reference to the attached drawings. The disclosure may, however, be embodied in many different forms and should not be construed as being limited to the embodiments of the disclosure set forth herein. In the drawings, parts not related to the disclosure are not illustrated for clarity of explanation, and like reference numerals denote like elements throughout.

FIG. 1 is a diagram for describing a general air conditioner control system.

In general, when a user 10 comes indoors after outdoor activities, the user 10 may feel very hot due to a high active metabolic rate. In this case, the user 10 may operate a general air conditioner 100 by setting a desired temperature of the air conditioner 100 to the lowest temperature, e.g., 18° C. When a time T has elapsed after the air conditioner 100 starts to operate, the active metabolic rate of the user 10 is stabled and the user 10 feels cold. In this case, the user 10 may turn off the air conditioner 100 or increase the set temperature of the air conditioner 100 to 25° C. However, the user 10 has already felt cold and thus experiences discomfort due to overcooling.

Discomfort experienced by the user 10 with the general air conditioner control system will now be described in more detail with reference to graphs 101 and 102 of FIG. 1.

Referring to a comfortable temperature graph 101 indicating a set of temperatures at which the user 10 feels comfortable, the user 10 feels hot at first and thus feels comfortable at a low temperature, e.g., c ° C. However, because heat escapes from the body of the user 10 over time, the temperature at which the user 10 feels comfortable is gradually increased. After the temperature at which the user 10 feels comfortable is increased to a specific temperature, the temperature at which the user 10 feels comfortable is maintained at the specific temperature.

Meanwhile, referring to an indoor temperature change graph 102, when the user 10 feels hot at first and thus inputs 18° C. as the set temperature, an indoor temperature is gradually reduced. At a timing when 20 minutes have elapsed after the air conditioner 100 starts to operate, the user 10 may feel cold and change the set temperature to 25° C. In this case, the indoor temperature is not immediately

increased due to thermal inertia, but is further reduced during a time r and then is increased after the time r has elapsed. When the comfortable temperature graph **101** is compared to the indoor temperature change graph **102**, although the user **10** feels cold and changes the set temperature to 25° C. after 20 minutes, overcooling that makes the user **10** feel cold already occurs from a timing i before the user **10** changes the set temperature. Overcooling may mean that an indoor temperature is reduced below a comfortable temperature of the user **10**. Therefore, from the timing i after the user **10** operates the air conditioner **100** to a timing j when the indoor temperature becomes similar to the comfortable temperature, the user **10** continuously feels cold and experiences discomfort due to overcooling.

As such, according to the general air conditioner control system for controlling the set temperature through training merely based on a history of temperatures set by the user **10**, the set temperature is controlled after the user **10** already experiences discomfort, and a time for which the user **10** feels uncomfortable is increased. Therefore, a system for efficiently controlling the set temperature of the air conditioner **100** before the user **10** feels uncomfortable due to overcooling is required. The system for controlling the set temperature of the air conditioner **100** before the user **10** feels uncomfortable due to overcooling, by using artificial intelligence (AI) models, according to an embodiment of the disclosure, will now be described with reference to FIG. 2.

FIG. 2 is a diagram for describing an air conditioner control system according to an embodiment of the disclosure.

The air conditioner control system (hereinafter shortened to the control system) according to an embodiment of the disclosure may include an air conditioner **1000** and a server device **2000**. However, not all of the illustrated elements are required. The control system may be implemented with more or less elements than the illustrated elements. For example, the control system may further include a display device (e.g., a mobile device) connected to the server device **2000**. The display device (not shown) may be a device for executing a certain application provided by the server device **2000**, and displaying information provided by the server device **2000**, through an execution window of the certain application. The display device will be described in detail below with reference to FIG. 20. Each element of the control system according to an embodiment of the disclosure will now be described.

The air conditioner **1000** according to an embodiment of the disclosure may be a device for appropriately adjusting the temperature, humidity, quality, or flow of indoor air. The air conditioner **1000** may include a remote controller for controlling the air conditioner **1000**. The air conditioner **1000** may obtain indoor environment information by using at least one sensor. For example, the air conditioner **1000** may include a temperature sensor, a humidity sensor, and a dust sensor. The air conditioner **1000** may measure a current indoor temperature by using the temperature sensor, measure a current indoor humidity by using the humidity sensor, and measure a current indoor dust value by using the dust sensor.

According to an embodiment of the disclosure, the air conditioner **1000** may include a communication interface for communicating with an external device. For example, the air conditioner **1000** may communicate with the server device **2000** through the communication interface. The communication interface may include a short-range wireless communication interface and a mobile communication interface. The short-range wireless communication interface may

include a Bluetooth communication interface, a Bluetooth Low Energy (BLE) communication interface, a near field communication (NFC) interface, a wireless local area network (WLAN) (or Wi-Fi) communication interface, a Zigbee communication interface, an Infrared Data Association (IrDA) communication interface, a Wi-Fi Direct (WFD) communication interface, an ultra-wideband (UWB) communication interface, or an Ant+ communication interface, but is not limited thereto.

According to an embodiment of the disclosure, the air conditioner **1000** may upload the indoor environment information to the server device **2000** through the communication interface. The air conditioner **1000** may transmit the indoor environment information to the server device **2000** periodically or when a specific event occurs (e.g., when a request is received from the server device **2000**). According to an embodiment of the disclosure, the air conditioner **1000** may transmit a current set temperature, a current fan speed, and device information (e.g., device performance and depreciation), or an inquiry about settings of the air conditioner **1000** to the server device **2000** through the communication interface. The air conditioner **1000** may receive a request related to temperature adjustment or a request related to fan speed adjustment from the server device **2000** in response to the inquiry. For example, the air conditioner **1000** may receive a set temperature maintenance request, a set temperature reduction request, a set temperature increase request, a fan speed reduction request, a fan speed increase request, or a fan speed maintenance request from the server device **2000**, but is not limited thereto. To prevent overcooling, a case when the air conditioner **1000** receives a set temperature increase request or a fan speed reduction request will be described below as a representative example.

According to an embodiment of the disclosure, when a request related to set temperature adjustment is received from the server device **2000**, the air conditioner **1000** may adjust a set temperature. When a request related to fan speed adjustment is received from the server device **2000**, the air conditioner **1000** may adjust a fan speed.

According to an embodiment of the disclosure, the air conditioner **1000** may be embedded with an AI processor. The AI processor may be produced in the form of a dedicated hardware chip for AI or as a part of a general-purpose processor (e.g., a central processing unit (CPU) or an application processor) or a dedicated graphics processor (e.g., a graphics processing unit (GPU)), and be embedded in the air conditioner **1000**. The air conditioner **1000** may adjust the set temperature or the fan speed by using the AI processor. The air conditioner **1000** may recognize voice of the user **10** and execute a command corresponding to the voice of the user **10**, by using the AI processor.

According to an embodiment of the disclosure, the air conditioner **1000** may receive an analog voice signal through a microphone, and convert the voice signal into computer-readable text by using an automatic speech recognition (ASR) model. The air conditioner **1000** may identify the intention of utterance of the user **10** by analyzing the converted text by using a natural language understanding (NLU) model. Herein, the ASR model or the NLU model may be an AI model. The AI model may be processed by a dedicated AI processor designed with a hardware structure specialized for processing AI models. The AI model may be made through training. For example, a basic AI model may be trained based on a plurality of pieces of training data by using a learning algorithm, and thus an AI model for performing a specific function may be made. The AI model may include a plurality of neural network layers. Each of the

plurality of neural network layers has a plurality of weight values and performs neural network computation through computation between a computation result of a previous layer and the plurality of weight values.

Linguistic understanding is a technology for recognizing and applying/processing human languages/characters, and includes natural language processing, machine translation, dialog systems, question answering, speech recognition/synthesis, etc.

According to an embodiment of the disclosure, the air conditioner **1000** may obtain voice of the user **10** through the remote controller. When a voice signal of the user **10** is received through a microphone, the remote controller of the air conditioner **1000** may transmit the voice signal to the air conditioner **1000**. According to another embodiment of the disclosure, when the ASR model or the NLU model is driven by the server device **2000**, the remote controller of the air conditioner **1000** may transmit the voice signal to the server device **2000** and receive a voice recognition result from the server device **2000** through wireless communication (e.g., Wi-Fi). In this case, the remote controller of the air conditioner **1000** may transmit the voice recognition result to the air conditioner **1000**.

According to an embodiment of the disclosure, the air conditioner **1000** may provide various operation modes. For example, the air conditioner **1000** may provide an AI comfort mode, a cool mode, a dry mode, a purify mode, and a wind-free mode, but is not limited thereto. The AI comfort mode may be a mode for automatically optimizing settings of the air conditioner **1000** by using AI models. The cool mode may be a basic operation mode for cooling. The dry mode may be an operation mode for drying indoor air by sucking in moisture in the air. The purify mode may be an operation mode for making indoor air fresh and clean by filtering yellow dust or fine dust floating in the air. A case when the air conditioner **1000** operates in the AI comfort mode will be described below as an example.

The server device **2000** according to an embodiment of the disclosure may include an AI processor. The AI processor may generate a first AI model **201** for inferring a temperature at which the user **10** feels comfortable, and a second AI model **202** for predicting an indoor temperature, by training an artificial neural network (ANN). When the ANN is 'trained', it may mean that a mathematical model for allowing connections of neurons in the ANN to make optimal decisions by appropriately changing weights based on data is generated. The server device **2000** may store the first and second AI models **201** and **202** in a memory. The server device **2000** may obtain a comfortable temperature graph of the user **10** from the first AI model **201**, and obtain a predicted temperature graph from the second AI model **202**. The comfortable temperature graph may refer to a set of temperatures at which the user **10** feels comfortable based on an operation time of the air conditioner **1000**. The predicted temperature graph may refer to a set of temperatures predicted for future timings. The first and second AI models **201** and **202** will be described in detail below with reference to FIGS. 4 to 9.

According to an embodiment of the disclosure, the server device **2000** may be implemented as one server, two or more servers, or a cloud server. The server device **2000** may train the first and second AI models **201** and **202**, or receive and store the first and second AI models **201** and **202** trained by an external server.

According to an embodiment of the disclosure, the server device **2000** may include a communication interface for communicating with an external device. For example, when

the AI comfort mode is set, the server device **2000** may receive an inquiry about set temperature adjustment from the air conditioner **1000** and transmit a response to the inquiry to the air conditioner **1000**. For example, when the user **10** comes indoors from outside and inputs 'Too hot. AI comfort mode' by voice through the remote controller of the air conditioner **1000**, the air conditioner **1000** may start operating and set the operation mode to the AI comfort mode. In addition, the air conditioner **1000** may output a guidance message such as "You feel hot. Desired temperature is set to 18° C. for fast cooling and then will be adjusted before you feel cold" by voice or on a display in response to the user input. The air conditioner **1000** may inquire about set temperature adjustment by transmitting at least one of operation time information (e.g., an operation start timing), set temperature information (e.g., 18° C.), or current indoor temperature information (e.g., 30° C.) to the server device **2000** (operation S210). According to an embodiment of the disclosure, the air conditioner **1000** may repeatedly inquire of the server device **2000** about set temperature adjustment while the air conditioner **1000** is operating.

When the inquiry is received from the air conditioner **1000**, the server device **2000** may predict whether overcooling will occur, by using the first and second AI models **201** and **202** (operation S220). For example, the server device **2000** may obtain a comfortable temperature graph of the user **10** from the first AI model **201**, and obtain a predicted temperature graph from the second AI model **202** based on at least one of the operation time information of the air conditioner **1000** (e.g., a time from an operation start timing to a current timing), the set temperature information (e.g., 18° C.), or the current indoor temperature information (e.g., 30° C.). The server device **2000** may compare the comfortable temperature graph of the user **10** to the predicted temperature graph, and predict that overcooling will occur from a timing when the predicted temperature graph is reduced below the comfortable temperature graph of the user **10** (e.g., the timing i of FIG. 1). According to an embodiment of the disclosure, when it is predicted that overcooling will occur, the server device **2000** may predict changes in indoor temperature by applying one or more set temperatures to the second AI model **202** whenever the inquiry is received from the air conditioner **1000**. The server device **2000** may determine an appropriate set temperature for preventing overcooling, based on the predicted changes in indoor temperature. When the server device **2000** determines the appropriate set temperature (e.g., 25° C.) by using the second AI model **202**, the server device **2000** may transmit a set temperature increase request (e.g., a request instructing to increase from 18° C. to 25° C.) to the air conditioner **1000** (operation S230). The operation, performed by the server device **2000**, of predicting whether overcooling will occur will be described in detail below with reference to FIG. 3.

The air conditioner **1000** may adjust the set temperature based on the set temperature increase request received from the server device **2000**. For example, the air conditioner **1000** may change the set temperature from 18° C. to 25° C. at a timing when the set temperature increase request is received. According to an embodiment of the disclosure, the air conditioner **1000** may notify, by voice, that the set temperature has been adjusted. The air conditioner **1000** may change the set temperature output through the display, from 18° C. to 25° C.

Therefore, in the air conditioner control system according to an embodiment of the disclosure, the set temperature of the air conditioner **1000** may be appropriately adjusted using

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the first and second AI models **201** and **202** before the user **10** experiences discomfort due to overcooling. Furthermore, in the air conditioner control system according to an embodiment of the disclosure, wasted energy due to overcooling may be prevented and health problems due to overcooling (e.g., air-conditioning sickness) may also be prevented.

Meanwhile, according to another embodiment of the disclosure, the operation of the server device **2000** may be performed by the air conditioner **1000** or another indoor device (e.g., an air purifier, a refrigerator, a television (TV), a robot vacuum, or an AI speaker).

A method, performed by the air conditioner **1000**, of controlling a set temperature to prevent overcooling while operating in an AI comfort mode will now be described in detail with reference to FIG. 3.

FIG. 3 is a flowchart of an operating method of the air conditioner **1000**, according to an embodiment of the disclosure.

In operation **S310**, the air conditioner **1000** according to an embodiment of the disclosure may transmit current state information to the server device **2000**. In this case, the air conditioner **1000** may inquire of the server device **2000** about set temperature adjustment.

According to an embodiment of the disclosure, the current state information may include at least one of operation time information of the air conditioner **1000**, set temperature information of the air conditioner **1000**, or current indoor temperature information, but is not limited thereto. For example, the current state information may further include fan speed information, device performance information, and indoor humidity information.

The operation time information of the air conditioner **1000** may be information indicating a time elapsed from an operation start timing of the air conditioner **1000** to a current timing (e.g., 5 minutes, 10 minutes, or 20 minutes). The set temperature information of the air conditioner **1000** may be information about a desired temperature set in the air conditioner **1000** (e.g., 24° C.). The current indoor temperature information may include an indoor temperature value obtained using a temperature sensor of the air conditioner **1000** (e.g., 29° C.). The fan speed information may be information about a fan speed currently set in the air conditioner **1000** (e.g., Wind-free, Low, Medium, or High, or Level 0 (Wind-free), Level 1, Level 2, Level 3, Level 4, Level 5, or Max). The device performance information may be information about the performance of the air conditioner **1000**, and may include, for example, hardware performance information and decrepitude information, but is not limited thereto. The indoor humidity information may include an indoor humidity value obtained using a humidity sensor of the air conditioner **1000** (e.g., 50%), but is not limited thereto.

The air conditioner **1000** according to an embodiment of the disclosure may repeatedly inquire about set temperature adjustment by transmitting the current state information to the server device **2000** while operating in an AI comfort mode. For example, the air conditioner **1000** may inquire of the server device **2000** about set temperature adjustment in a certain cycle or at a timing designated by the server device **2000**.

In operation **S320**, the air conditioner **1000** according to an embodiment of the disclosure may receive a set temperature increase request from the server device **2000** in response to the current state information.

For example, when an overcooling period, in which a first predicted temperature graph obtained from the second AI

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model **202** based on the current state information is reduced below a comfortable temperature graph obtained from the first AI model **201**, is identified, the air conditioner **1000** may receive the set temperature increase request from the server device **2000**. According to an embodiment of the disclosure, the set temperature increase request may include a set temperature value. The set temperature increase request may include next inquiry timing information (e.g., after 5 minutes).

According to an embodiment of the disclosure, when the current state information is received from the air conditioner **1000**, the server device **2000** may obtain the comfortable temperature graph from the first AI model **201** and obtain the first predicted temperature graph from the second AI model **202** based on the current state information. The server device **2000** may identify the overcooling period by comparing the comfortable temperature graph to the first predicted temperature graph.

According to an embodiment of the disclosure, the comfortable temperature graph may indicate a set of temperatures at which a user feels comfortable based on an operation time of the air conditioner **1000**. The comfortable temperature graph may have a form of a solid line or a set of dots, but is not limited thereto.

According to an embodiment of the disclosure, the comfortable temperature graph may include an unstable period in which a comfortable temperature changes based on an operation time of the air conditioner **1000**, and a stable period in which the comfortable temperature is constantly maintained. For example, although the user feels hot and thus feels comfortable at a low temperature before the air conditioner **1000** starts to operate, heat escapes from the body of the user over time after the air conditioner **1000** starts to operate, and thus the temperature at which the user feels comfortable is gradually increased. When a certain time has elapsed after the air conditioner **1000** starts to operate, the user body or an indoor temperature may enter a thermal equilibrium state and the temperature at which the user feels comfortable is constantly maintained at a specific temperature.

According to an embodiment of the disclosure, the comfortable temperature graph may be obtained from the first AI model **201** for inferring a comfortable temperature of the user. The first AI model **201** may be a model configured to infer a comfortable temperature through training based on user settings information (e.g., a set temperature at a first timing and an operation time from an operation start timing to the first timing), environment information (e.g., an indoor temperature, an outdoor temperature, an indoor humidity, an outdoor humidity, an indoor-outdoor temperature difference, and an indoor-outdoor humidity difference), and device information (e.g., fan speed information). The first AI model **201** will be described in detail below with reference to FIG. 4.

According to an embodiment of the disclosure, the server device **2000** may obtain the comfortable temperature graph by applying at least one of operation timing information of the air conditioner **1000**, fan speed information of the air conditioner **1000**, indoor humidity information, outdoor humidity information, outdoor temperature information, an indoor-outdoor humidity difference, or an indoor-outdoor temperature difference to the first AI model **201**, but is not limited thereto. The operation timing information may be information about a time point or time period when the air conditioner **1000** is operating. For example, the operation timing information may include time period information

such as morning, afternoon, evening, night, noon, or dawn, and time point information such as a specific time indicated by a clock.

For example, the server device **2000** may obtain a set of temperatures at which the user feels comfortable based on an operation time of the air conditioner **1000** under a first condition indicating a ‘Low’ fan speed of the air conditioner **1000**, a time period of ‘evening’, and an indoor humidity of ‘60%’, as the comfortable temperature graph by using the first AI model **201**.

According to an embodiment of the disclosure, because a temperature at which the user feels comfortable varies depending on a time period, a fan speed, weather, or the like, the comfortable temperature graph obtained from the first AI model **201** may have various forms based on a time period, a fan speed, weather, an indoor humidity, an outdoor humidity, or the like.

According to an embodiment of the disclosure, the first predicted temperature graph may be obtained by applying, to the second AI model **202**, the current state information received from the air conditioner **1000**. The first predicted temperature graph may refer to a set of temperatures predicted based on the current state information. The first predicted temperature graph may have a form of a solid line or a set of dots, but is not limited thereto.

According to an embodiment of the disclosure, the first predicted temperature graph may be obtained by further inputting additional information to the second AI model **202** in addition to the current state information received from the air conditioner **1000**. For example, the first predicted temperature graph may be obtained by further inputting at least one of operation timing information of the air conditioner **1000**, indoor humidity information, outdoor temperature information, outdoor humidity information, weather information, performance information of the air conditioner **1000**, or installation space information of the air conditioner **1000** to the second AI model **202**. The performance information of the air conditioner **1000** may include hardware performance information, decrepitude information, and model information, but is not limited thereto. The installation space information of the air conditioner **1000** may include information about a thermal insulation level, information about whether a cooking appliance producing heat (e.g., a gas stove or an induction stove) is used, information about whether a lighting device producing heat is used, and information about whether windows are open or closed, but is not limited thereto.

According to an embodiment of the disclosure, the server device **2000** may obtain the additional information from an external server, home appliances, or a home network system. For example, the server device **2000** may obtain weather information, outdoor temperature information, and outdoor humidity information from a weather server. The server device **2000** may obtain indoor humidity information from an air purifier or a humidifier, and obtain, from a cooking appliance (e.g., a gas stove or an induction stove), information about whether the cooking appliance is used. The server device **2000** may obtain, from a home network system, information about whether a lighting device producing heat is used, and information about whether windows are open or closed.

The server device **2000** may compare the comfortable temperature graph obtained from the first AI model **201**, to the first predicted temperature graph obtained from the second AI model **202** based on the current state information. The server device **2000** may identify that an overcooling period in which the first predicted temperature graph is

reduced below the comfortable temperature graph will occur at a future timing, based on the comparison result.

When the overcooling period is identified, the server device **2000** may generate a set temperature increase request including information instructing to change the set temperature of the air conditioner **1000** to a first temperature. According to an embodiment of the disclosure, the first temperature may be determined from among one or more set temperatures, based on a result of comparing the comfortable temperature graph to one or more second predicted temperature graphs obtained by inputting the one or more set temperatures to the second AI model **202**. Herein, the one or more set temperatures may be temperatures higher than a current set temperature of the air conditioner **1000**.

For example, the server device **2000** may obtain the one or more second predicted temperature graphs by applying the one or more set temperatures to the second AI model **202**. In this case, the server device **2000** may identify a second predicted temperature graph which is not reduced below the comfortable temperature graph and converges on the comfortable temperature graph, from among the one or more second predicted temperature graphs. The server device **2000** may determine a set temperature corresponding to the identified second predicted temperature graph, as the first temperature. For example, the server device **2000** may obtain a plurality of second predicted temperature graphs by individually applying set temperatures such as 23° C., 24° C., and 25° C. to the second AI model **202**. In this case, when second-1 and second-2 predicted temperature graphs respectively corresponding to 23° C. and 24° C. exhibit an overcooling period reduced below the comfortable temperature graph, the server device **2000** may not select 23° C. and 24° C. as the set temperature. When a second-3 predicted temperature graph corresponding to 25° C. is not reduced below the comfortable temperature graph and converges on the comfortable temperature graph, the server device **2000** may determine 25° C. as the set temperature. The operation, performed by the server device **2000**, of determining the set temperature will be described in detail below with reference to FIG. 11.

The server device **2000** may transmit, to the air conditioner **1000**, the set temperature increase request instructing to change the set temperature to the first temperature (e.g., 25° C.). The server device **2000** may request the air conditioner **1000** to inquire about set temperature adjustment again after a certain time has elapsed.

In operation S330, the air conditioner **1000** according to an embodiment of the disclosure may adjust the set temperature of the air conditioner **1000** based on the set temperature increase request. For example, when the set temperature increase request includes information instructing to change the set temperature (or desired temperature) to the first temperature (e.g., 25° C.), the air conditioner **1000** may adjust the set temperature to the first temperature (e.g., 25° C.).

Meanwhile, the air conditioner **1000** may receive a request instructing to inquire about set temperature adjustment again after a certain time has elapsed, from the server device **2000** together with the set temperature increase request. In this case, when the certain time designated by the server device **2000** has elapsed, the air conditioner **1000** may inquire about set temperature adjustment again by transmitting the current state information to the server device **2000**.

According to an embodiment of the disclosure, the air conditioner **1000** may control the set temperature by using the first and second AI models **201** and **202** driven by the

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server device **2000**, in such a manner that the user always feels comfortable without feeling cold.

The first AI model **201** for inferring a comfortable temperature of a user will now be described in more detail with reference to FIG. 4.

FIG. 4 is a diagram for describing an operation of training the first AI model **201**, according to an embodiment of the disclosure.

According to an embodiment of the disclosure, an AI processor included in the server device **2000** may generate a comfortable temperature inference model by training an ANN. Herein, the comfortable temperature inference model may be the first AI model **201**. When the ANN is ‘trained’, it may mean that a mathematical model for allowing connections of neurons in the ANN to make optimal decisions by appropriately changing weights based on training data **400** is generated. For example, the AI processor may configure the first AI model **201** for inferring the comfortable temperature, by obtaining the training data **400** and training a basic machine learning algorithm by using the training data **400**.

According to an embodiment of the disclosure, the training data **400** may include condition information and comfortable temperature information. The condition information may include user settings information (e.g., a desired temperature set by a user and a timing when the user sets the desired temperature), environment information (e.g., an indoor temperature, an indoor humidity, an outdoor temperature, an outdoor humidity, an indoor-outdoor humidity difference, an indoor-outdoor temperature difference, and weather), and device information (e.g., a fan speed currently set in the air conditioner **1000**), but is not limited thereto. For example, the condition information may further include time period information (e.g., morning, afternoon, evening, or dawn), season information, and activity information of the user (e.g., information about whether the user is exercising or cooking). The comfortable temperature information may include a temperature at which the user feels comfortable under each condition. For example, when the user changes the desired temperature from 18° C. to 25° C. at a first timing when 20 minutes have elapsed after the air conditioner **1000** starts to operate, when a fan speed at the first timing is ‘Low’, and when an indoor temperature is 20° C., it means that the user feels cold at 20° C. at the first timing, and thus 22° C. obtained by adding 2° C. to the indoor temperature may serve as the comfortable temperature of the first timing. Therefore, the training data **400** in this case may include {Set timing: 20 minutes after starting operation, Set temperature: 25° C., Current indoor temperature: 20° C., Current indoor humidity: 60%, Fan speed: Low, Comfortable temperature: 22° C.}.

According to an embodiment of the disclosure, the training data **400** may be obtained from the air conditioner **1000**, an external server, or at least one home appliance. For example, the server device **2000** may obtain user settings information, fan speed information, indoor temperature information, and indoor humidity information from the air conditioner **1000**. The server device **2000** may obtain weather information, outdoor humidity information, and outdoor temperature information from the external server (e.g., a weather server). The server device **2000** may obtain indoor temperature information and indoor humidity information from a humidifier or an air purifier. The server device **2000** may obtain, from a robot vacuum, an image including an activity of the user. The server device **2000** may obtain

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exercise information of the user from an exercise machine and obtain, from a cooking appliance, information about whether the user is cooking.

According to an embodiment of the disclosure, the basic machine learning algorithm may be a linear regression algorithm or a deep learning algorithm, but is not limited thereto. The linear regression algorithm may model a linear correlation between a dependent variable y and one or more independent variables (or explanatory variables) X. The deep learning algorithm is a machine learning algorithm that attempts high-level abstraction (i.e., summarizing key contents or functions in large amounts of data or complex data) through a combination of several nonlinear transformation methods.

According to an embodiment of the disclosure, the first AI model **201** may be trained using the training data **400** to generate a comfortable temperature database (DB) **410**. According to an embodiment of the disclosure, the first AI model **201** may update the comfortable temperature DB **410** when the training data **400** is added or updated. According to an embodiment of the disclosure, the comfortable temperature DB **410** may be a DB storing comfortable temperature graphs of the user under various conditions, and include a table in which the condition information is mapped to the comfortable temperature graphs of the user. The condition information may include time information (e.g., operation timing information of the air conditioner **1000**), environment information (e.g., an indoor humidity, an outdoor humidity, an indoor temperature, an outdoor temperature, an indoor-outdoor humidity difference, an indoor-outdoor temperature difference, and weather), and device information (e.g., fan speed information and performance information), but is not limited thereto.

According to an embodiment of the disclosure, the comfortable temperature DB **410** may be generated while the first AI model **201** is being trained or after the first AI model **201** is trained, but is not limited thereto.

According to an embodiment of the disclosure, the comfortable temperature DB **410** may store various comfortable temperature graphs under various conditions. The various comfortable temperature graphs will be described in detail below with reference to FIGS. 7A and 7B.

Meanwhile, according to an embodiment of the disclosure, the first AI model **201** may be configured to generate a comfortable temperature graph including an unstable period and a stable period. An operation, performed by the first AI model **201**, of modeling an unstable period will now be described with reference to FIG. 5, and then an operation, performed by the first AI model **201**, of modeling a stable period will be described with reference to FIG. 6.

FIG. 5 is a diagram for describing an operation of modeling an unstable period of a comfortable temperature graph, according to an embodiment of the disclosure.

Referring to FIG. 5, an AI processor may model a comfortable temperature of an unstable period of a comfortable temperature graph by using a linear regression algorithm (also referred to as a linear regression AI model). For example, when a user increases a set temperature of the air conditioner **1000** at a first timing, the AI processor may define “Indoor temperature at first timing +1° C. or +2° C.” as a comfortable temperature of the first timing. The AI processor may store the comfortable temperature of the first timing and an operation time of the air conditioner **1000** at the first timing as first training data. When two or more comfortable temperatures are stored as training data, the linear regression algorithm (or linear regression AI model)

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may be trained based on the training data to infer a comfortable temperature of the air conditioner **1000** based on an operation time.

For example, the linear regression algorithm (or linear regression AI model) may estimate c and Δ by finding a linear equation which is closest to all points. That is, the linear regression algorithm (or linear regression AI model) may obtain a linear equation **500** of Comfortable temperature $y=x*\Delta+c$. Meanwhile, although the first AI model **201** is described as a linear regression AI model in FIG. 5, but is not limited thereto and may be a non-linear AI model.

FIG. 6 is a diagram for describing an unstable period and a stable period of a comfortable temperature graph, according to an embodiment of the disclosure.

The first AI model **201** may determine whether a comfortable temperature enters a stable state through thermal equilibrium of the body of a user or an indoor space. The stable state refers to a state in which the comfortable temperature does not change and is constantly maintained based on an operation time of the air conditioner **1000**. For example, the first AI model **201** may be trained based on training data to determine that the comfortable temperature enters the stable state, when a certain time (e.g., 30 minutes) has elapsed after the air conditioner **1000** starts to operate, when a set temperature of the air conditioner **1000** does not change during a certain time h , and when an indoor temperature is maintained in a certain range. In this case, the first AI model **201** may define the indoor temperature during the certain time h as the comfortable temperature of the stable period.

Referring to FIG. 6, the first AI model **201** may identify a coordinate (T, b) at which a first graph **601** corresponding to the linear equation **500** of FIG. 5 meets a second graph **602** obtained by connecting indoor temperatures during the certain time h . The first AI model **201** may generate a comfortable temperature graph **600** to follow the first graph **601** before a timing T **603** and follow the second graph **602** after the timing T **603**. For example, the first AI model **201** may obtain, as an output value, the comfortable temperature graph **600** including an unstable period **610** in which the comfortable temperature changes based on an operation time of the air conditioner **1000** before the timing T **603**, and a stable period **620** in which the comfortable temperature is constantly maintained after the timing T **603**.

In general, although it is very hot, when about 30 minutes have elapsed after the air conditioner **1000** is turned on, it enters a stable state through thermal equilibrium. Therefore, when the air conditioner **1000** is turned on, a pattern like the comfortable temperature graph **600** of FIG. 6 may always exist.

FIG. 7A is a diagram for describing a plurality of comfortable temperature graphs obtained by applying a plurality of fan speed values to the first AI model **201**, according to an embodiment of the disclosure.

Referring to FIG. 7A, the first AI model **201** may generate various comfortable temperature graphs based on fan speed values of the air conditioner **1000**. For example, when a fan speed is 'Low', the first AI model **201** may obtain a first comfortable temperature graph **701** in which a comfortable temperature is increased from $a^{\circ}\text{C.}$ to $b^{\circ}\text{C.}$ in an unstable period **710** and is maintained at $b^{\circ}\text{C.}$ in a stable period **720**.

When the fan speed is 'Medium', the first AI model **201** may obtain a second comfortable temperature graph **702** in which the comfortable temperature is increased from $a+1^{\circ}\text{C.}$ to $b+1^{\circ}\text{C.}$ in the unstable period **710** and is maintained at $b+1^{\circ}\text{C.}$ in the stable period **720**. When the fan speed is 'High', the first AI model **201** may obtain a third comfort-

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able temperature graph **703** in which the comfortable temperature is increased from $a+2^{\circ}\text{C.}$ to $b+2^{\circ}\text{C.}$ in the unstable period **710** and is maintained at $b+2^{\circ}\text{C.}$ in the stable period **720**. When the fan speed is 'Wind-free', the first AI model **201** may obtain a fourth comfortable temperature graph **704** in which the comfortable temperature is increased from $a-1^{\circ}\text{C.}$ to $b-1^{\circ}\text{C.}$ in the unstable period **710** and is maintained at $b-1^{\circ}\text{C.}$ in the stable period **720**.

According to an embodiment of the disclosure, the comfortable temperature is increased when the fan speed of the air conditioner **1000** is increased, and is reduced when the fan speed is reduced. According to an embodiment of the disclosure, the first to fourth comfortable temperature graphs **701** to **704** may be stored in the comfortable temperature DB **410**.

FIG. 7B is a diagram for describing a plurality of comfortable temperature graphs obtained by applying a plurality of time periods to the first AI model **201**, according to an embodiment of the disclosure.

Referring to FIG. 7B, the first AI model **201** may generate various comfortable temperature graphs based on a time point (or time period) when the air conditioner **1000** operates. For example, when the air conditioner **1000** operates around midday, the first AI model **201** may obtain a fifth comfortable temperature graph **705** in which a comfortable temperature is increased from $a^{\circ}\text{C.}$ to $b^{\circ}\text{C.}$ in the unstable period **710** and is maintained at $b^{\circ}\text{C.}$ in the stable period **720**.

When the air conditioner **1000** operates in the evening, the first AI model **201** may obtain a sixth comfortable temperature graph **706** in which the comfortable temperature is increased from $a+1^{\circ}\text{C.}$ to $b+1^{\circ}\text{C.}$ in the unstable period **710** and is maintained at $b+1^{\circ}\text{C.}$ in the stable period **720**. When the air conditioner **1000** operates in the morning, the first AI model **201** may obtain a seventh comfortable temperature graph **707** in which the comfortable temperature is increased from $a+1.5^{\circ}\text{C.}$ to $b+1.5^{\circ}\text{C.}$ in the unstable period **710** and is maintained at $b+1.5^{\circ}\text{C.}$ in the stable period **720**. When the air conditioner **1000** operates at dawn, the first AI model **201** may obtain an eighth comfortable temperature graph **708** in which the comfortable temperature is increased from $a+2^{\circ}\text{C.}$ to $b+2^{\circ}\text{C.}$ in the unstable period **710** and is maintained at $b+2^{\circ}\text{C.}$ in the stable period **720**.

According to an embodiment of the disclosure, the comfortable temperature may be reduced when the air conditioner **1000** operates around midday, but be increased when the air conditioner **1000** operates in the morning or at dawn. According to an embodiment of the disclosure, the fifth to eighth comfortable temperature graphs **705** to **708** may be stored in the comfortable temperature DB **410**.

Meanwhile, according to an embodiment of the disclosure, the comfortable temperature graph may also vary depending on activity information of a user. For example, the comfortable temperature may be reduced when the user is exercising or has just finished exercising, and be increased when the user reads a book on a desk or watches TV on a sofa. According to an embodiment of the disclosure, the comfortable temperature DB **410** may store various comfortable temperature graphs based on the activity information of the user.

FIG. 8 is a diagram for describing an operation of obtaining a comfortable temperature graph from the first AI model **201**, according to an embodiment of the disclosure.

Referring to FIG. 8, according to an embodiment of the disclosure, the server device **2000** may obtain a comfortable temperature graph **800** by applying at least one of time information **801**, environment information **802**, or device

information **803** to the first AI model **201**, but is not limited thereto. The time information **801** may include operation timing information of the air conditioner **1000** (e.g., an operation time point or time period). The environment information **802** may include an indoor humidity, an outdoor humidity, an indoor temperature, an outdoor temperature, an indoor-outdoor temperature difference, an indoor-outdoor humidity difference, and weather. The device information **803** may include fan speed information, device performance information, and device decrepitude information.

According to an embodiment of the disclosure, the server device **2000** may obtain the comfortable temperature graph **800** as an output value of the first AI model **201**. For example, the first AI model **201** may generate the comfortable temperature graph **800** as the output value based on the time information **801**, the environment information **802**, and the device information **803**. The first AI model **201** may transmit the comfortable temperature graph **800** to a processor of the server device **2000**. According to an embodiment of the disclosure, the first AI model **201** may extract the comfortable temperature graph **800** from the comfortable temperature DB **410**, based on the time information **801**, the environment information **802**, and the device information **803**. In this case, the comfortable temperature graph **800** may include an unstable period in which a comfortable temperature is increased based on an operation time of the air conditioner **1000**, and a stable period in which the comfortable temperature is constantly maintained.

According to an embodiment of the disclosure, the first AI model **201** may obtain the comfortable temperature graph **800** as the output value by further applying context information of a user. For example, when the server device **2000** collects the context information of the user (e.g., "Back home after run") from a mobile device (e.g., a wearable device) of the user, the first AI model **201** may output the comfortable temperature graph **800** to which the context information of the user is reflected. When the user comes back home after run, the comfortable temperature graph **800** may be generated to be lower than that of a general case.

When the server device **2000** collects, from a cooking appliance (e.g., an induction stove), information indicating that the user is using the cooking appliance, the first AI model **201** may output the comfortable temperature graph **800** to which context information indicating that the user is cooking is reflected.

FIG. 9 is a diagram for describing an operation of obtaining a predicted temperature graph from the second AI model **202**, according to an embodiment of the disclosure.

According to an embodiment of the disclosure, an AI processor included in the server device **2000** may generate an indoor temperature prediction model by training an ANN. Herein, the indoor temperature prediction model may be the second AI model **202**. When the ANN is 'trained', it may mean that a mathematical model for allowing connections of neurons in the ANN to make optimal decisions by appropriately changing weights based on training data is generated. Herein, the ANN may include at least one of a deep neural network (DNN), a recurrent neural network (RNN), a convolutional neural network (CNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), or a deep Q-network (DQN), but is not limited thereto.

For example, the AI processor may configure the second AI model **202** for predicting indoor temperatures of future timings, by obtaining training data and training an ANN model by using the training data. According to an embodiment of the disclosure, the training data may include user

settings information (e.g., a desired temperature set by a user and a timing when the user sets the desired temperature), environment information (e.g., an indoor temperature, an indoor humidity, an outdoor temperature, an outdoor humidity, an indoor-outdoor humidity difference, an indoor-outdoor temperature difference, and weather), device information (e.g., a fan speed currently set in the air conditioner **1000**, and device performance), and space information (e.g., thermal insulation information, information about whether a lighting device is used, and information about whether windows are open or closed). For example, the user may change the desired temperature from 18° C. to 25° C. at a first timing when 20 minutes have elapsed after the air conditioner **1000** starts to operate, a fan speed at the first timing may be 'Low', and an indoor temperature at the first timing may be 20° C. An indoor temperature at a second timing when 5 minutes have elapsed from the first timing may be 21° C., and an indoor temperature at a third timing when 10 minutes have elapsed from the first timing may be 22° C. In this case, the training data may include {Set timing: First timing (20 minutes after starting operation), Set temperature at first timing: 25° C., Fan speed at first timing: Low, Indoor temperature at first timing: 20° C., Indoor temperature at second timing: 21° C., Indoor temperature at third timing: 22° C.}. The accuracy of temperature values predicted by the second AI model **202** may be increased when the second AI model **202** is trained based on various types of training data.

Referring to FIG. 9, the server device **2000** may obtain a predicted temperature graph **900** by applying time information **901**, environment information **902**, device information **903**, and space information **904** to the second AI model **202**. According to an embodiment of the disclosure, not all of the time information **901**, the environment information **902**, the device information **903**, and the space information **904** may be applied to the second AI model **202**, and only some of them may be applied. The time information **901** may include operation time information of the air conditioner **1000**, and operation timing information of the air conditioner **1000** (e.g., an operation time point or time period). The environment information **902** may include an indoor humidity, an outdoor humidity, an indoor temperature, an outdoor temperature, an indoor-outdoor temperature difference, an indoor-outdoor humidity difference, and weather. The device information **903** may include fan speed information set in the air conditioner **1000**, set temperature information of the air conditioner **1000**, device performance information, and device decrepitude information, but is not limited thereto. The space information **904** may include information about a thermal insulation level, information about whether a cooking appliance producing heat (e.g., a gas stove or an induction stove) is used, information about whether a lighting device producing heat is used, and information about whether windows are open or closed, but is not limited thereto.

According to an embodiment of the disclosure, the server device **2000** may obtain the predicted temperature graph **900** as an output value of the second AI model **202**. For example, when a current indoor temperature is 30° C., and when a current set temperature of the air conditioner **1000** is 18° C., the second AI model **202** may generate, as the output value, the predicted temperature graph **900** in which the temperature is reduced from 30° C. to 18° C. over time.

According to an embodiment of the disclosure, the predicted temperature graph **900** may be generated in various forms based on a set temperature of the air conditioner **1000**, a fan speed, an operation time of the air conditioner **1000**,

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and a current indoor temperature. For example, when the fan speed is High, compared to a case when the fan speed is Low, a temperature may be reduced rapidly.

An operation, performed by the server device **2000**, of identifying whether overcooling will occur, based on a comfortable temperature graph obtained from the first AI model **201** and a predicted temperature graph obtained from the second AI model **202** will now be described in detail with reference to FIG. **10**.

FIG. **10** is a diagram for describing an operation, performed by the server device **2000**, of identifying an overcooling period, according to an embodiment of the disclosure.

According to an embodiment of the disclosure, the air conditioner **1000** may inquire of the server device **2000** about set temperature adjustment at a timing t_0 . In this case, the air conditioner **1000** may transmit, to the server device **2000**, current state information including a set temperature at the timing t_0 (e.g., 18° C.), an operation time until the timing t_0 (e.g., 6 minutes), and an indoor temperature at the timing t_0 (e.g., 27° C.).

When the set temperature adjustment inquiry is received from the air conditioner **1000**, the server device **2000** may obtain a comfortable temperature graph **1001** from the first AI model **201**, and obtain a first predicted temperature graph **1002** from the second AI model **202**. For example, the server device **2000** may obtain the comfortable temperature graph **1001** by applying at least one of time information (e.g., the timing t_0 =midday), environment information (e.g., an indoor temperature at the timing t_0 , an indoor humidity at the timing t_0 , weather at the timing t_0 , an outdoor temperature at the timing t_0 , and an outdoor humidity at the timing t_0), or device information (e.g., a fan speed at the timing t_0) to the first AI model **201**, but is not limited thereto. The comfortable temperature graph **1001** may include an unstable period and a stable period. The operation, performed by the server device **2000**, of obtaining the comfortable temperature graph **1001** is described above in relation to FIG. **8**, and thus a detailed description thereof is not provided herein.

The server device **2000** may obtain the first predicted temperature graph **1002** by applying, to the second AI model **202**, the current state information received from the air conditioner **1000**. For example, the server device **2000** may obtain the first predicted temperature graph **1002** by applying the set temperature at the timing t_0 (e.g., 18° C.), the operation time until the timing t_0 (e.g., 6 minutes), and the indoor temperature at the timing t_0 (e.g., 27° C.) to the second AI model **202**. Because the current indoor temperature is 27° C. and the current set temperature of the air conditioner **1000** is 18° C., the second AI model **202** may generate, as an output value, the first predicted temperature graph **1002** in which the temperature is reduced from 27° C. to 18° C. over time. The operation, performed by the server device **2000**, of obtaining the first predicted temperature graph **1002** is described above in relation to FIG. **9**, and thus a detailed description thereof is not provided herein.

According to an embodiment of the disclosure, the server device **2000** may identify an overcooling period by comparing the comfortable temperature graph **1001** to the first predicted temperature graph **1002**. For example, because the comfortable temperature graph **1001** is reduced below the first predicted temperature graph **1002** at a timing t_1 , the server device **2000** may predict that overcooling will occur from the timing t_1 .

According to an embodiment of the disclosure, when it is predicted that overcooling will occur, the server device **2000** may increase the set temperature of the air conditioner **1000**

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to a specific temperature to prevent overcooling. The operation, performed by the server device **2000**, of adjusting the set temperature of the air conditioner **1000** to the specific temperature will now be described in detail with reference to FIG. **11**.

FIG. **11** is a diagram for describing an operation, performed by the server device **2000**, of determining a set temperature of the air conditioner **1000** by using the second AI model **202**, according to an embodiment of the disclosure.

Referring to FIG. **11**, to prevent overcooling, the server device **2000** may apply a plurality of set temperatures to the second AI model **202** at a timing t_0 . For example, the server device **2000** may apply 24.5° C., 25° C., and 25.5° C. to the second AI model **202**. In this case, the server device **2000** may obtain, from the second AI model **202**, a predicted temperature graph corresponding to each set temperature. Even when the set temperature is changed at the timing t_0 , an indoor temperature is not immediately increased due to thermal inertia. Therefore, the predicted temperature graph corresponding to each set temperature may have a pattern in which the indoor temperature is reduced during a certain time (e.g., r) from the timing t_0 and then is increased after the certain time (e.g., r) has elapsed.

The server device **2000** may obtain a second-1 predicted temperature graph **1101** corresponding to the set temperature of 24.5° C., a second-2 predicted temperature graph **1102** corresponding to the set temperature of 25.0° C., and a second-3 predicted temperature graph **1103** corresponding to the set temperature of 25.5° C. In this case, the second-1 predicted temperature graph **1101** corresponding to the set temperature of 24.5° C. shows an overcooling period in which a predicted indoor temperature is reduced below a comfortable temperature from a timing t_2 , and thus the server device **2000** may not select 24.5° C. as the set temperature of the air conditioner **1000**. When the second-2 predicted temperature graph **1102** corresponding to the set temperature of 25.0° C. is compared to the second-3 predicted temperature graph **1103** corresponding to the set temperature of 25.5° C., the second-2 predicted temperature graph **1102** is closest to the comfortable temperature graph **1001**. That is, the second-2 predicted temperature graph **1102** may not be reduced below the comfortable temperature graph **1001** and converge on the comfortable temperature graph **1001**. Therefore, the server device **2000** may select 25.0° C. as the set temperature of the air conditioner **1000**.

Because the set temperature of the air conditioner **1000** is determined as 25.0° C. by using the second AI model **202**, the server device **2000** may transmit, to the air conditioner **1000**, a set temperature increase request instructing to change the set temperature to 25° C. The air conditioner **1000** may change the set temperature from 18° C. to 25° C. based on the set temperature increase request. In this case, because the predicted indoor temperature is not reduced below the comfortable temperature, a user may not experience discomfort due to overcooling. The method, performed by the server device **2000**, of adjusting the set temperature of the air conditioner **1000** will now be described in more detail with reference to FIG. **12**.

FIG. **12** is a flowchart of a method of adjusting a set temperature of the air conditioner **1000**, according to an embodiment of the disclosure.

In operation S1201, the air conditioner **1000** according to an embodiment of the disclosure may obtain current state information. For example, the air conditioner **1000** may obtain at least one of operation time information (e.g., time information from an operation start timing to a current

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timing), set temperature information of the air conditioner **1000**, or current indoor temperature information, but is not limited thereto. For example, the air conditioner **1000** may further obtain fan speed information, device performance information, and indoor humidity information as the current state information.

In operation **S1202**, the air conditioner **1000** according to an embodiment of the disclosure may inquire of the server device **2000** about set temperature adjustment. In this case, the air conditioner **1000** may transmit the current state information to the server device **2000**.

In operation **S1203**, when the inquiry is received from the air conditioner **1000**, the server device **2000** according to an embodiment of the disclosure may obtain a comfortable temperature graph from the first AI model **201**. In this case, the comfortable temperature graph may include an unstable period in which a comfortable temperature is increased based on an operation time of the air conditioner **1000**, and a stable period in which the comfortable temperature is constantly maintained. The operation, performed by the server device **2000**, of obtaining the comfortable temperature graph from the first AI model **201** is described in detail above in relation to FIG. **8**, and thus a detailed description thereof is not provided herein.

In operation **S1204**, the server device **2000** according to an embodiment of the disclosure may obtain a first predicted temperature graph based on the current state information from the second AI model **202**. For example, the server device **2000** may obtain the first predicted temperature graph by applying at least one of the operation time information of the air conditioner **1000** (e.g., time information from an operation start timing to a current timing), the set temperature information of the air conditioner **1000**, or the current indoor temperature information to the second AI model **202**. The operation, performed by the server device **2000**, of obtaining the first predicted temperature graph is described in detail above in relation to FIG. **9**, and thus a detailed description thereof is not provided herein.

In operation **S1205**, the server device **2000** according to an embodiment of the disclosure may identify whether an overcooling period will occur. The server device **2000** may identify the overcooling period by comparing the comfortable temperature graph to the first predicted temperature graph. For example, the server device **2000** may identify a specific timing when the first predicted temperature graph is reduced below the comfortable temperature graph, and determine that overcooling will occur from the specific timing.

In operation **S1206**, when the overcooling period is not identified, the set temperature of the air conditioner **1000** at a current timing does not need to be adjusted, and thus the server device **2000** according to an embodiment of the disclosure may request the air conditioner **1000** to maintain the current set temperature and to inquire again after a certain time has elapsed. For example, the server device **2000** may request the air conditioner **1000** to inquire about set temperature adjustment again after 10 minutes.

Herein, the certain time may be arbitrarily designated by the server device **2000**. For example, the server device **2000** may designate the certain time as a short time when an indoor temperature changes rapidly, and designate the certain time as a long time when the indoor temperature changes gently. For example, the server device **2000** may designate the certain time within 5 minutes when the temperature change is greater than a first threshold value, designate the certain time between 10 minutes and 30 minutes when the temperature change is between the first

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threshold value and a second threshold value, and designate the certain time as 30 minutes or more when the temperature change is less than the second threshold value.

In operation **S1207**, when the overcooling period is identified, the server device **2000** according to an embodiment of the disclosure may obtain a plurality of second predicted temperature graphs by applying a plurality of set temperatures to the second AI model **202**. Because the indoor temperature changes based on the set temperature of the air conditioner **1000**, the plurality of second predicted temperature graphs may have various forms.

In operation **S1208**, the server device **2000** according to an embodiment of the disclosure may compare the plurality of second predicted temperature graphs to the comfortable temperature graph. For example, the server device **2000** may identify a predicted temperature graph which is not reduced below the comfortable temperature graph, from among the plurality of second predicted temperature graphs.

In operation **S1209**, the server device **2000** according to an embodiment of the disclosure may determine whether the set temperature of the air conditioner **1000** currently needs to be controlled, based on the result of comparing the plurality of second predicted temperature graphs to the comfortable temperature graph.

For example, when the predicted temperature graph is increased above the comfortable temperature graph by more than a threshold value because the set temperature of the air conditioner **1000** is currently controlled, the server device **2000** may determine that control is not required at the current timing. When the predicted temperature graph is higher than the comfortable temperature graph by more than the threshold value, it may mean that the indoor temperature is not sufficiently reduced to a temperature at which a user feels comfortable. Therefore, when the set temperature is currently controlled, overcooling will not occur but the user may feel hot, and thus the server device **2000** may determine that the set temperature does not need to be controlled at the current timing.

Meanwhile, when a predicted temperature graph obtained by applying a first temperature to the second AI model **202** is close to the comfortable temperature graph, the server device **2000** may determine that the set temperature of the air conditioner **1000** needs to be controlled to the first temperature at the current timing.

In operation **S1210**, when it is determined that the set temperature of the air conditioner **1000** does not need to be currently controlled, the server device **2000** according to an embodiment of the disclosure may request the air conditioner **1000** to maintain the current set temperature and to inquire again after a certain time has elapsed. Herein, the certain time may be arbitrarily designated by the server device **2000**. For example, the server device **2000** may designate the certain time as a short time when an indoor temperature changes rapidly, and designate the certain time as a long time when the indoor temperature changes gently.

When it is determined in operation **S1211** that the certain time requested by the server device **2000** has elapsed, the air conditioner **1000** according to an embodiment of the disclosure may return to operation **S1201** and obtain the current state information again and inquire of the server device **2000** about set temperature adjustment again.

In operation **S1212**, when it is determined that the set temperature of the air conditioner **1000** needs to be controlled, the server device **2000** according to an embodiment of the disclosure may request the air conditioner **1000** to increase the set temperature and to inquire again after a certain time has elapsed.

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For example, when a predicted temperature graph obtained by applying a first temperature to the second AI model **202** is close to the comfortable temperature graph, the server device **2000** may transmit, to the air conditioner **1000**, a request instructing to adjust the set temperature of the air conditioner **1000** to the first temperature. The server device **2000** may request the air conditioner **1000** to inquire about set temperature adjustment again after 20 minutes.

In operation **S1213**, the air conditioner **1000** according to an embodiment of the disclosure may adjust the set temperature based on the set temperature increase request. For example, the air conditioner **1000** may adjust the set temperature to the first temperature.

When it is determined in operation **S1214** that the certain time requested by the server device **2000** has elapsed, the air conditioner **1000** according to an embodiment of the disclosure may return to operation **S1201** and obtain the current state information again and inquire of the server device **2000** about set temperature adjustment again.

According to an embodiment of the disclosure, because the state of a space where the user is located continuously changes, the server device **2000** may control the set temperature of the air conditioner **1000** at appropriate timings to make the user continuously feel comfortable, by continuously monitoring the state of the space around the user. The operation, performed by the server device **2000**, of controlling the set temperature of the air conditioner **1000** by continuously monitoring the state of the space around the user will now be described in more detail with reference to FIGS. **13** and **14**.

FIG. **13** is a diagram for describing an operation, performed by the server device **2000**, of requesting the air conditioner **1000** to inquire again after a certain time has elapsed, according to an embodiment of the disclosure.

Referring to FIG. **13**, the air conditioner **1000** may inquire of the server device **2000** about set temperature adjustment at a timing when 5 minutes have elapsed after starting operation (operation **S1310**). In this case, the air conditioner **1000** may transmit, to the server device **2000**, current state information including operation time information (e.g., 5 minutes), a current indoor temperature (e.g., 26° C.), and a current set temperature (e.g., 18° C.).

The server device **2000** may compare a comfortable temperature graph **1311** obtained from the first AI model **201**, to a first predicted temperature graph **1312** obtained by applying the current state information to the second AI model **202**. The server device **2000** may predict that overcooling will occur from a timing when 20 minutes have elapsed after the air conditioner **1000** starts to operate, based on the comparison result. In this case, the server device **2000** may obtain a plurality of second predicted temperature graphs **1301**, **1302**, and **1303** by applying a plurality of set temperatures to the second AI model **202**. For example, the server device **2000** may obtain a second-1 predicted temperature graph **1301** by applying a set temperature of 25.5° C. to the second AI model **202**, obtain a second-2 predicted temperature graph **1302** by applying a set temperature of 25.0° C. to the second AI model **202**, and obtain a second-3 predicted temperature graph **1303** by applying a set temperature of 24.5° C. to the second AI model **202**.

Because all of the plurality of second predicted temperature graphs **1301**, **1302**, and **1303** are higher than the comfortable temperature graph **1311** by more than a threshold temperature **K** (e.g., 2° C.) at a timing T_k , the server device **2000** may determine that the set temperature of the air conditioner **1000** does not need to be currently controlled. When the server device **2000** adjusts the set tem-

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perature of the air conditioner **1000** at a current timing (e.g., 5 minutes after the air conditioner **1000** starts to operate), an indoor temperature is not sufficiently reduced to a temperature at which a user feels comfortable, and thus the server device **2000** may determine not to adjust the set temperature of the air conditioner **1000** at the current timing.

The server device **2000** may request the air conditioner **1000** to maintain the current set temperature and to inquire about set temperature adjustment again after 5 minutes (operation **S1320**). In this case, the air conditioner **1000** may maintain the current set temperature. When a certain time requested by the server device **2000** (e.g., 5 minutes) has elapsed, the air conditioner **1000** may inquire of the server device **2000** about set temperature adjustment again. A detailed description thereof will now be provided with reference to FIG. **14**.

FIG. **14** is a diagram for describing an operation, performed by the air conditioner **1000**, of inquiring of the server device **2000** about set temperature adjustment again after a certain time has elapsed, according to an embodiment of the disclosure.

Referring to FIG. **14**, when a certain time requested by the server device **2000** (e.g., 5 minutes) has elapsed, the air conditioner **1000** may inquire of the server device **2000** about set temperature adjustment (operation **S1410**). For example, because the server device **2000** has requested to inquire about set temperature adjustment again after 5 minutes at the timing when 5 minutes have elapsed after the air conditioner **1000** starts to operate, the air conditioner **1000** may inquire of the server device **2000** about set temperature adjustment at a timing when 10 minutes have elapsed after starting operation. In this case, the air conditioner **1000** may transmit, to the server device **2000**, current state information including operation time information (e.g., 10 minutes), a current indoor temperature (e.g., 22° C.), and a current set temperature (e.g., 18° C.).

The server device **2000** may compare a comfortable temperature graph **1411** obtained from the first AI model **201**, to a first predicted temperature graph **1412** obtained by applying the current state information to the second AI model **202**. The server device **2000** may predict that overcooling will occur from a timing when 20 minutes have elapsed after the air conditioner **1000** starts to operate, based on the comparison result. In this case, the server device **2000** may obtain a plurality of second predicted temperature graphs **1401**, **1402**, and **1403** by applying a plurality of set temperatures to the second AI model **202**. For example, the server device **2000** may obtain a second-1 predicted temperature graph **1401** by applying a set temperature of 25.5° C. to the second AI model **202**, obtain a second-2 predicted temperature graph **1402** by applying a set temperature of 25.0° C. to the second AI model **202**, and obtain a second-3 predicted temperature graph **1403** by applying a set temperature of 24.5° C. to the second AI model **202**.

Because differences between the comfortable temperature graph **1411** at a timing T_k and the plurality of second predicted temperature graphs **1401**, **1402**, and **1403** at the timing T_k are less than a threshold temperature **K** (e.g., 2° C.), the server device **2000** may determine that the set temperature of the air conditioner **1000** needs to be controlled at a current timing (e.g., 10 minutes after starting operation). The second-1 predicted temperature graph **1401** corresponding to the set temperature of 24.5° C. shows an overcooling period in which a predicted indoor temperature is reduced below a comfortable temperature from a timing T_2 , and thus the server device **2000** may not select 24.5° C. as the set temperature of the air conditioner **1000**. When the

second-2 predicted temperature graph **1402** corresponding to the set temperature of 25.0° C. is compared to the second-3 predicted temperature graph **1403** corresponding to the set temperature of 25.5° C., the second-2 predicted temperature graph **1402** is closest to the comfortable temperature graph **1411**. The second-2 predicted temperature graph **1402** may not be reduced below the comfortable temperature graph **1411** and converge on the comfortable temperature graph **1411**. Therefore, the server device **2000** may select 25.0° C. as the set temperature of the air conditioner **1000**.

Because the set temperature of the air conditioner **1000** is determined as 25.0° C. by using the second AI model **202**, the server device **2000** may transmit, to the air conditioner **1000**, a set temperature increase request instructing to change the set temperature to 25° C. (operation **S1420**). In this case, the server device **2000** may request the air conditioner **1000** to inquire about set temperature adjustment again after 10 minutes.

The air conditioner **1000** may change the set temperature from 18° C. to 25° C. at 10 minutes after starting operation, based on the set temperature increase request. In this case, because the predicted indoor temperature is close to the comfortable temperature, a user may continuously feel comfortable. When 10 minutes have elapsed after the air conditioner **1000** changes the set temperature to 25° C., the air conditioner **1000** may inquire of the server device **2000** about set temperature adjustment again. Meanwhile, the air conditioner **1000** may receive the set temperature increase request from the server device **2000** a plurality of times. The operation, performed by the air conditioner **1000**, of receiving the set temperature increase request from the server device **2000** a plurality of times will now be described in detail with reference to FIGS. **15A** and **15B**.

FIGS. **15A** and **15B** are diagrams for describing an operation, performed by the air conditioner **1000**, of adjusting a set temperature a plurality of times, according to an embodiment of the disclosure.

Referring to FIG. **15A**, the air conditioner **1000** may start to operate at a timing T_0 and adjust a set temperature to 18° C. The air conditioner **1000** may inquire of the server device **2000** about set temperature adjustment at a timing T_1 when 6 minutes have elapsed after starting operation, and receive, from the server device **2000**, a set temperature increase request instructing to change the set temperature to 23° C. The air conditioner **1000** may receive, from the server device **2000**, a request instructing to inquire about set temperature adjustment again after 9 minutes. The air conditioner **1000** may change the set temperature from 18° C. to 23° C. at the timing T_1 (e.g., 6 minutes after starting operation) based on the set temperature increase request. Even when the air conditioner **1000** changes the set temperature to 23° C. at the timing T_1 , an indoor temperature is not immediately increased due to thermal inertia, but is further reduced during a time r and then is increased after the time r has elapsed.

The air conditioner **1000** may inquire of the server device **2000** about set temperature adjustment again at a current timing T_2 (e.g., 15 minutes after starting operation) when 9 minutes have elapsed from the timing T_1 (e.g., 6 minutes after starting operation) (operation **S1510**). For example, the air conditioner **1000** may inquire about set temperature adjustment by transmitting, to the server device **2000**, current state information including operation time information at the current timing T_2 (e.g., 15 minutes after starting operation), indoor temperature information at the current

timing T_2 (e.g., 20° C.), and set temperature information at the current timing T_2 (e.g., 23° C.).

When the set temperature adjustment inquiry is received, the server device **2000** may identify an overcooling period by comparing a comfortable temperature graph **1501** obtained from the first AI model **201**, to a first predicted temperature graph **1502** obtained from the second AI model **202** based on the current state information (operation **S1520**). For example, the server device **2000** may determine that overcooling will occur from a timing T_3 when the first predicted temperature graph **1502** is reduced below the comfortable temperature graph **1501**. In this case, the server device **2000** may determine a set temperature to be set at the current timing T_2 , by using the second AI model **202**.

Referring to FIG. **15B**, the server device **2000** may obtain one or more second predicted temperature graphs by applying one or more set temperatures to the second AI model **202**. The server device **2000** may determine 25° C. from among the one or more set temperatures, based on the one or more second predicted temperature graphs. In this case, a predicted temperature graph **1503** corresponding to the set temperature of 25° C. may not be reduced below the comfortable temperature graph **1501** and be closest to the comfortable temperature graph **1501**.

The server device **2000** may transmit, to the air conditioner **1000**, a request instructing to increase the set temperature to 25° C. (operation **S1530**). The air conditioner **1000** may change the set temperature from 23° C. to 25° C. at the timing T_2 . In this case, overcooling may not occur at the timing T_3 and thus a user may continuously feel comfortable.

According to an embodiment of the disclosure, the air conditioner **1000** may gradually change the set temperature from 18° C. to 23° C. at the timing T_1 and from 23° C. to 25° C. at the timing T_2 in association with the server device **2000**, thereby preventing the user from experiencing discomfort due to overcooling. The set temperature of the air conditioner **1000** is changed twice in FIGS. **15A** and **15B**, but is not limited thereto. The air conditioner **1000** may change the set temperature three or more times in association with the server device **2000**.

Meanwhile, the air conditioner **1000** may adjust a fan speed to prevent overcooling. The method, performed by the air conditioner **1000**, of adjusting a fan speed will now be described in detail with reference to FIG. **16**.

FIG. **16** is a flowchart of a method, performed by the air conditioner **1000**, of adjusting a fan speed, according to an embodiment of the disclosure.

In operation **S1610**, the air conditioner **1000** according to an embodiment of the disclosure may transmit current state information to the server device **2000**. The air conditioner **1000** may inquire of the server device **2000** about fan speed adjustment.

According to an embodiment of the disclosure, the current state information may include at least one of operation time information of the air conditioner **1000**, set temperature information of the air conditioner **1000**, current indoor temperature information, or fan speed information, but is not limited thereto. For example, the current state information may further include device performance information and indoor humidity information.

The air conditioner **1000** according to an embodiment of the disclosure may repeatedly inquire about fan speed adjustment by transmitting the current state information to the server device **2000** while operating in an AI comfort mode. For example, the air conditioner **1000** may inquire of

the server device **2000** about fan speed adjustment in a certain cycle or at a timing designated by the server device **2000**.

In operation **S1620**, the air conditioner **1000** according to an embodiment of the disclosure may receive a fan speed reduction request from the server device **2000** in response to the current state information.

For example, when an overcooling period, in which a first predicted temperature graph obtained from the second AI model **202** based on the current state information is reduced below a comfortable temperature graph obtained from the first AI model **201** based on the fan speed information, is identified, the air conditioner **1000** may receive the fan speed reduction request from the server device **2000**. According to an embodiment of the disclosure, the fan speed reduction request may include a fan speed value. The fan speed reduction request may include next inquiry timing information (e.g., after 5 minutes).

According to an embodiment of the disclosure, when the current state information is received from the air conditioner **1000**, the server device **2000** may obtain the comfortable temperature graph from the first AI model **201**. The comfortable temperature graph may indicate a set of temperatures at which a user feels comfortable based on an operation time of the air conditioner **1000**. The comfortable temperature graph may vary depending on the current fan speed. Therefore, the server device **2000** may obtain the comfortable temperature graph by applying, to the first AI model **201**, the fan speed information included in the current state information. The server device **2000** may obtain the first predicted temperature graph by applying, to the second AI model **202**, the current state information received from the air conditioner **1000**. The server device **2000** may identify that an overcooling period in which the first predicted temperature graph is reduced below the comfortable temperature graph will occur at a future timing, by comparing the comfortable temperature graph to the first predicted temperature graph.

When the overcooling period is identified, the server device **2000** may generate a fan speed reduction request including information instructing to change the fan speed of the air conditioner **1000** to a first level. In this case, the first level may be determined by applying one or more fan speeds to the first and second AI models **201** and **202**. For example, the server device **2000** may obtain one or more second comfortable temperature graphs by applying the one or more fan speeds to the first AI model **201**. The server device **2000** may obtain one or more second predicted temperature graphs by applying the one or more fan speeds to the second AI model **202**. The server device **2000** may determine the fan speed of the first level from among the one or more fan speeds, based on a result of comparing the one or more second comfortable temperature graphs to the one or more second predicted temperature graphs. The second predicted temperature graph obtained by applying the fan speed of the first level to the second AI model **202** may not be reduced below the second comfortable temperature graph obtained by applying the fan speed of the first level to the first AI model **201**, and converge on the second comfortable temperature graph. Herein, the first level may be lower than the fan speed currently set in the air conditioner **1000**.

The server device **2000** may transmit, to the air conditioner **1000**, the fan speed reduction request instructing to adjust the fan speed to the first level. The server device **2000** may request the air conditioner **1000** to inquire about fan speed adjustment again after a certain time has elapsed.

In operation **S1630**, the air conditioner **1000** according to an embodiment of the disclosure may adjust the fan speed of the air conditioner **1000** based on the fan speed reduction request. For example, when the fan speed reduction request includes information instructing to change the fan speed to the first level (e.g., Low), and when the current fan speed is 'High', the air conditioner **1000** may adjust the fan speed from 'High' to 'Low'.

Meanwhile, the air conditioner **1000** may receive a request instructing to inquire about fan speed adjustment again after a certain time has elapsed, from the server device **2000** together with the fan speed reduction request. In this case, when the certain time designated by the server device **2000** has elapsed, the air conditioner **1000** may inquire about fan speed adjustment again by transmitting the current state information to the server device **2000**.

According to an embodiment of the disclosure, the air conditioner **1000** may control the fan speed by using the first and second AI models **201** and **202** driven by the server device **2000**, in such a manner that the user always feels comfortable without feeling cold. The operation, performed by the air conditioner **1000**, of adjusting the fan speed in association with the server device **2000** will now be described in more detail with reference to FIG. 17.

FIG. 17 is a diagram for describing an operation, performed by the server device **2000**, of determining a fan speed of the air conditioner **1000** by using the first AI model **201** and the second AI model **202**, according to an embodiment of the disclosure.

Referring to FIG. 17, the air conditioner **1000** may inquire of the server device **2000** about fan speed adjustment at a timing T_0 when 10 minutes have elapsed after starting operation (operation **S1710**). In this case, the air conditioner **1000** may transmit, to the server device **2000**, current state information including operation time information (e.g., 10 minutes), a current indoor temperature (e.g., 24° C.), a current set temperature (e.g., 18° C.), and a current fan speed (e.g., High).

The server device **2000** may compare a first comfortable temperature graph **1701** obtained by applying the current fan speed (e.g., High) to the first AI model **201**, to a first predicted temperature graph **1711** obtained by applying the current state information to the second AI model **202**. The server device **2000** may predict that overcooling will occur from a timing T_1 , based on the comparison result.

When it is predicted that overcooling will occur, the server device **2000** may obtain a plurality of comfortable temperature graphs **1702**, **1703**, and **1704** by applying a plurality of fan speeds (e.g., Medium, Low, and Wind-free) to the first AI model **201**. For example, the server device **2000** may obtain a second comfortable temperature graph **1702** by applying the 'Fan speed: Medium' to the first AI model **201**, obtain a third comfortable temperature graph **1703** by applying the 'Fan speed: Low' to the first AI model **201**, and obtain a fourth comfortable temperature graph **1704** by applying the 'Fan speed: Wind-free' to the first AI model **201**. The second comfortable temperature graph **1702** may be 1° C. lower than the first comfortable temperature graph **1701**, the third comfortable temperature graph **1703** may be 2° C. lower than the first comfortable temperature graph **1701**, and the fourth comfortable temperature graph **1704** may be 3° C. lower than the first comfortable temperature graph **1701**. That is, the comfortable temperature graph may be reduced when the fan speed is reduced.

Meanwhile, the server device **2000** may obtain a plurality of predicted temperature graphs **1712**, **1713**, and **1714** by applying the plurality of fan speeds (e.g., Medium, Low, and

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Wind-free) to the second AI model 202. For example, the server device 2000 may obtain a second predicted temperature graph 1712 by applying the 'Fan speed: Medium' to the second AI model 202, obtain a third predicted temperature graph 1713 by applying the 'Fan speed: Low' to the second AI model 202, and obtain a fourth predicted temperature graph 1714 by applying the 'Fan speed: Wind-free' to the second AI model 202. A temperature indicated by the predicted temperature graph may be reduced gently when the fan speed is reduced.

The server device 2000 may determine the fan speed as 'Medium' by comparing the plurality of comfortable temperature graphs 1701, 1702, 1703, and 1704 to the plurality of predicted temperature graphs 1711, 1712, 1713, and 1714. The second predicted temperature graph 1712 obtained by applying the 'Fan speed: Medium' to the second AI model 202 is not reduced below the second comfortable temperature graph 1702 obtained by applying the 'Fan speed: Medium' to the first AI model 201, and converges on the second comfortable temperature graph 1702. Therefore, when the fan speed of the air conditioner 1000 is adjusted from High to Medium, discomfort of a user due to overcooling may be prevented.

The server device 2000 may transmit, to the air conditioner 1000, a fan speed reduction request instructing to adjust the fan speed to Medium (operation S1720) and, in this case, the server device 2000 may request the air conditioner 1000 to inquire about fan speed adjustment again after 10 minutes.

The air conditioner 1000 may change the fan speed from High to Medium based on the fan speed reduction request. In this case, because the predicted indoor temperature is close to the comfortable temperature, the user may continuously feel comfortable. When 10 minutes have elapsed after the air conditioner 1000 changes the fan speed to Medium, the air conditioner 1000 may inquire of the server device 2000 about fan speed adjustment again.

Meanwhile, according to an embodiment of the disclosure, the air conditioner 1000 may receive the fan speed reduction request from the server device 2000 a plurality of times. For example, the air conditioner 1000 may receive, from the server device 2000 at certain time intervals, a first fan speed reduction request instructing to change the fan speed from High to Medium, a second fan speed reduction request instructing to change the fan speed from Medium to Low, and a third fan speed reduction request instructing to change the fan speed from Low to Wind-free.

According to an embodiment of the disclosure, the air conditioner 1000 may receive a set temperature increase request and a fan speed reduction request at the same time from the server device 2000. The operation, performed by the air conditioner 1000, of receiving the set temperature increase request and the fan speed reduction request at the same time from the server device 2000 will now be described in detail with reference to FIG. 18.

FIG. 18 is a flowchart of a method, performed by the air conditioner 1000, of adjusting a set temperature and a fan speed, according to an embodiment of the disclosure.

In operation S1810, the air conditioner 1000 according to an embodiment of the disclosure may transmit current state information to the server device 2000. The air conditioner 1000 may inquire of the server device 2000 about settings.

According to an embodiment of the disclosure, the current state information may include at least one of operation time information of the air conditioner 1000, set temperature information of the air conditioner 1000, current indoor temperature information, or fan speed information, but is not

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limited thereto. For example, the current state information may further include device performance information and indoor humidity information.

The air conditioner 1000 according to an embodiment of the disclosure may repeatedly inquire about settings by transmitting the current state information to the server device 2000 while operating in an AI comfort mode. For example, the air conditioner 1000 may inquire of the server device 2000 about settings in a certain cycle or at a timing designated by the server device 2000.

In operation S1820, the air conditioner 1000 according to an embodiment of the disclosure may receive a set temperature increase request and a fan speed reduction request from the server device 2000.

For example, when an overcooling period, in which a first predicted temperature graph obtained from the second AI model 202 based on the current state information is reduced below a first comfortable temperature graph obtained from the first AI model 201 based on the fan speed information, is identified, the air conditioner 1000 may receive the set temperature increase request and the fan speed reduction request from the server device 2000. According to an embodiment of the disclosure, the set temperature increase request may include a set temperature value. According to an embodiment of the disclosure, the fan speed reduction request may include a fan speed value. The set temperature increase request and the fan speed reduction request may include next inquiry timing information (e.g., after 5 minutes).

According to an embodiment of the disclosure, when the current state information is received from the air conditioner 1000, the server device 2000 may obtain the first comfortable temperature graph from the first AI model 201. The first comfortable temperature graph may indicate a set of temperatures at which a user feels comfortable based on an operation time of the air conditioner 1000. The first comfortable temperature graph may vary depending on the current fan speed. Therefore, the server device 2000 may obtain the first comfortable temperature graph by applying, to the first AI model 201, the fan speed information included in the current state information. The server device 2000 may obtain the first predicted temperature graph by applying, to the second AI model 202, the current state information received from the air conditioner 1000. The server device 2000 may identify that an overcooling period in which the first predicted temperature graph is reduced below the first comfortable temperature graph will occur at a future timing.

When the overcooling period is identified, the server device 2000 may generate a request including information instructing to change the set temperature of the air conditioner 1000 to a first temperature and change the fan speed of the air conditioner 1000 to a first level.

According to an embodiment of the disclosure, the server device 2000 may determine the first temperature from among one or more set temperatures and determine the first level from among one or more fan speeds, based on a result of comparing one or more second predicted temperature graphs obtained by applying the one or more set temperatures and the one or more fan speeds to the second AI model 202, to one or more second comfortable temperature graphs obtained by applying the one or more fan speeds to the first AI model 201. Herein, the one or more set temperatures may be higher than the current set temperature of the air conditioner 1000, and the one or more fan speeds may be lower than the current fan speed. A predicted temperature graph obtained by applying the fan speed of the first level and the first temperature to the second AI model 202 may not be

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reduced below a comfortable temperature graph obtained by applying the fan speed of the first level to the first AI model 201, and converge on the comfortable temperature graph.

The server device 2000 may transmit, to the air conditioner 1000, the request instructing to adjust the fan speed to the first level and to adjust the set temperature to the first temperature. The server device 2000 may request the air conditioner 1000 to inquire about settings adjustment again after a certain time has elapsed.

In operation S1830, the air conditioner 1000 according to an embodiment of the disclosure may adjust the set temperature and the fan speed of the air conditioner 1000 based on the set temperature increase request and the fan speed reduction request.

For example, when the set temperature increase request includes information instructing to change the set temperature to the first temperature (e.g., 25° C.), and when the current set temperature is 18° C., the air conditioner 1000 may change the set temperature from 18° C. to 25° C. When the fan speed reduction request includes information instructing to change the fan speed to the first level (e.g., Low), and when the current fan speed is 'High', the air conditioner 1000 may adjust the fan speed from 'High' to 'Low'.

Meanwhile, the air conditioner 1000 may receive a request instructing to inquire about settings adjustment again after a certain time has elapsed, from the server device 2000 together with the set temperature increase request and the fan speed reduction request. In this case, when the certain time designated by the server device 2000 has elapsed, the air conditioner 1000 may inquire about settings adjustment again by transmitting the current state information to the server device 2000.

According to an embodiment of the disclosure, the air conditioner 1000 may control the set temperature and the fan speed by using the first and second AI models 201 and 202 driven by the server device 2000, in such a manner that the user always feels comfortable without feeling cold. The operation, performed by the air conditioner 1000, of adjusting the set temperature and the fan speed in association with the server device 2000 will now be described in more detail with reference to FIGS. 19A and 19B.

FIGS. 19A and 19B are diagrams for describing an operation, performed by the server device 2000, of determining a set temperature and a fan speed, according to an embodiment of the disclosure.

Referring to FIG. 19A, the air conditioner 1000 may start to operate at a timing T_0 and adjust a set temperature to 18° C. The air conditioner 1000 may inquire of the server device 2000 about settings adjustment at a timing T_1 when 6 minutes have elapsed after starting operation, and receive, from the server device 2000, a set temperature increase request instructing to change the set temperature to 23° C. The air conditioner 1000 may receive, from the server device 2000, a request instructing to inquire about settings adjustment again after 9 minutes. The air conditioner 1000 may change the set temperature from 18° C. to 23° C. at the timing T_1 (e.g., 6 minutes after starting operation) based on the set temperature increase request. Even when the air conditioner 1000 changes the set temperature to 23° C. at the timing T_1 , an indoor temperature is not immediately increased due to thermal inertia, but is further reduced during a time r and then is increased after the time r has elapsed.

The air conditioner 1000 may inquire of the server device 2000 about settings adjustment again at a current timing T_2 (e.g., 15 minutes after starting operation) when 9 minutes

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have elapsed from the timing T_1 (e.g., 6 minutes after starting operation) (operation S1910). For example, the air conditioner 1000 may inquire about settings adjustment by transmitting, to the server device 2000, current state information including operation time information at the current timing T_2 (e.g., 15 minutes after starting operation), indoor temperature information at the current timing T_2 (e.g., 20° C.), set temperature information at the current timing T_2 (e.g., 23° C.), and fan speed information at the current timing T_2 (e.g., Fan speed: Medium).

When the settings adjustment inquiry is received, the server device 2000 may identify an overcooling period by comparing a first comfortable temperature graph 1901 obtained from the first AI model 201, to a first predicted temperature graph 1902 obtained from the second AI model 202 (operation S1920). For example, the server device 2000 may determine that overcooling will occur from a timing T_3 when the first predicted temperature graph 1902 is reduced below the first comfortable temperature graph 1901. In this case, the server device 2000 may determine a set temperature and a fan speed to be set at the current timing T_2 , by using the first and second AI models 201 and 202.

Referring to FIG. 19B, the server device 2000 may obtain one or more second predicted temperature graphs by applying one or more set temperatures and one or more fan speeds to the second AI model 202. The server device 2000 may obtain one or more second comfortable temperature graphs by applying the one or more fan speeds to the first AI model 201. The server device 2000 may determine 24° C. from among the one or more set temperatures and determine 'Low' from among the one or more fan speeds, based on the one or more second predicted temperature graphs and the one or more second comfortable temperature graphs. In this case, a second predicted temperature graph 1904 corresponding to the set temperature of 24° C. and the fan speed of 'Low' may not be reduced below a second comfortable temperature graph 1903 and be closest to the second comfortable temperature graph 1903. According to an embodiment of the disclosure, the second comfortable temperature graph 1903 corresponding to the fan speed of 'Low' may be 1° C. lower than the first comfortable temperature graph 1901 corresponding to the fan speed of 'Medium'.

The server device 2000 may transmit, to the air conditioner 1000, a request instructing to increase the set temperature to 24° C. and to reduce the fan speed to 'Low' (operation S1930). The air conditioner 1000 may change the set temperature from 23° C. to 24° C. and change the fan speed from 'Medium' to 'Low' at the timing T_2 . In this case, overcooling may not occur at the timing T_3 and thus a user may continuously feel comfortable.

According to an embodiment of the disclosure, the air conditioner 1000 may adjust the set temperature and the fan speed in association with the server device 2000, thereby preventing the user from experiencing discomfort due to overcooling.

Meanwhile, the server device 2000 may provide information related to the air conditioner 1000 to the user through a display device connected to the server device 2000. The operation, performed by the server device 2000, of providing the information related to the air conditioner 1000 to the user will now be described with reference to FIG. 20.

FIG. 20 is a diagram for describing an operation, performed by the server device 2000, of providing energy saving information based on an AI comfort mode through a display device 3000, according to an embodiment of the disclosure. FIG. 20 shows, as an example, a case when the display device 3000 executes a specific application provided

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by the server device **2000**. The specific application may be a computer program for providing information related to home appliances.

Referring to reference numeral **2000-1** of FIG. **20**, the display device **3000** may execute the specific application and display a list of home appliances of a user. The home appliances of the user may be connected to the server device **2000** by using the same account. For example, when the user executes the specific application for controlling the home appliances through the display device **3000**, the display device **3000** may display icons indicating the air conditioner **1000**, a refrigerator, a TV, and an air purifier on an execution screen of the application. In this case, the display device **3000** may receive a user input for selecting an icon **2001** indicating the air conditioner **1000**.

According to an embodiment of the disclosure, the display device **3000** may be a device for executing the specific application provided by the server device **2000**. The display device **3000** may be a mobile device of the user, a wearable device of the user, a refrigerator including a display, a TV, a desktop computer, or a laptop computer, but is not limited thereto. For convenience of explanation, the display device **3000** is assumed as a mobile device in FIG. **20**.

Referring to reference numeral **2000-2** of FIG. **20**, the display device **3000** may display an interface screen for providing information about the air conditioner **1000**, based on the user input for selecting the icon **2001** indicating the air conditioner **1000**. In this case, a current operation mode of the air conditioner **1000** may be displayed on the interface screen. For example, when the air conditioner **1000** is currently operating in an AI comfort mode, an image indicating the AI comfort mode and text such as 'AI comfort' may be displayed at an upper part of the screen.

When the user desires to monitor energy consumption of the air conditioner **1000**, the display device **3000** may receive a user input for selecting an energy monitor icon **2002**.

Referring to reference numeral **2000-3** of FIG. **20**, the display device **3000** may provide energy information related to the AI comfort mode in response to the user input for selecting the energy monitor icon **2002**. For example, the display device **3000** may provide information about energy saved in the AI comfort mode compared to a normal mode (e.g., "AI comfort mode has saved 256 kWh"). The display device **3000** may provide a first image **2004** comparatively showing a (predicted) indoor temperature graph and a comfortable temperature graph corresponding to the normal mode, and a second image **2005** comparatively showing a (predicted) indoor temperature graph and a comfortable temperature graph corresponding to the AI comfort mode. The first image **2004** related to the normal mode may show that an overcooling period will occur.

The display device **3000** may provide information **2006** including energy saved in the AI comfort mode compared to the normal mode (e.g., 256 kWh until today), and an estimated cost (e.g., □37,481).

FIGS. **21** and **22** are block diagrams for describing functions of the air conditioner **1000**, according to an embodiment of the disclosure.

As illustrated in FIG. **21**, the air conditioner **1000** according to an embodiment of the disclosure may include a processor **1400**, a communication interface **1500**, and a memory **1800**. However, not all of the illustrated elements are required. The air conditioner **1000** may be implemented with more or less elements than the illustrated elements. As illustrated in FIG. **22**, the air conditioner **1000** according to an embodiment of the disclosure may include a blower **1100**,

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a filter **1200**, an outputter **1300**, the processor **1400**, the communication interface **1500**, a sensor **1600**, a user inputter **1700**, and the memory **1800**.

The elements of the air conditioner **1000** will now be described one by one.

The blower **1100** may include an air intake **1110**, a blowing fan **1120**, a motor **1130**, and an air discharger **1140**, but is not limited thereto. The air intake **1110** may suck in the air around the air conditioner **1000**.

The blowing fan **1120** may allow external air to enter the air conditioner **1000** through the air intake **1110** by forming an air flow. Meanwhile, the blowing fan **1120** may also allow the air purified by the filter **1200** to be discharged from the air conditioner **1000** through the air discharger **1140**. The blowing fan **1120** may be rotated by the motor **1130** to form an air flow. A rotational speed (i.e., revolutions per minute) of the motor **1130** may be adjusted under the control by the processor **1400**.

The blowing fan **1120** may include a high-pass (whirlwind) fan and a circulator fan. Depending on implementation, a plurality of high-pass fans and a plurality of circulator fans may be provided. The high-pass fan may blow out fast and strong cold air. For example, the high-pass fan may blow out strong cold air in a high-pass manner as soon as the air is sucked in. The circulator fan may send wind far away. The circulator fan may form a strong jet stream at a side wind door (e.g., a circular air outlet) with the wind sucked in.

The air discharger **1140** may include a metal cooling panel for discharging the cold air, and the circular air outlet. The metal cooling panel may include 1 mm sand-sized micro-holes for sending out the cold air. The cold air may be uniformly spread out through the metal cooling panel including the micro-holes.

The filter **1200** may include at least one filter and be mounted in a case body. The filter **1200** may be located adjacent to the air intake **1110** of the air conditioner **1000** to filter out pollutants included in the air entering through the air intake **1110**. According to an embodiment of the disclosure, the filter **1200** may include a plurality of filters. For example, the filter **1200** may include a prefilter **1210**, a high-efficiency particulate air (HEPA) filter **1220**, and a deodorizing filter **1230**, but is not limited thereto. For example, the filter **1200** may further include various types of functional filters.

The prefilter **1210** may be a filter for removing relatively large particles. For example, the prefilter **1210** may remove large dust, hair, and pet hair. The HEPA filter **1220** may remove fine dust, mites, viruses, and various bacteria such as fungi. The deodorizing filter **1230** may remove various odors and harmful gases in an indoor environment. The deodorizing filter **1230** may also be referred to as a carbon filter. The functional filters (not shown) may be for removing pollen, dust mites, bacteria, etc. The functional filters (not shown) may include an antibacterial filter and a carbon filter.

The outputter **1300** is for outputting an audio signal or a video signal. The outputter **1300** may include a display **1310** and an audio outputter **1320**.

When the display **1310** is layered with a touchpad to configure a touchscreen, the display **1310** may be used not only as an output device but also as an input device. The display **1310** may include at least one of a liquid crystal display (LCD), a thin film transistor-liquid crystal display (TFT-LCD), an organic light-emitting diode (OLED), a flexible display, a three-dimensional (3D) display, or an electrophoretic display. Depending on implementation of the air conditioner **1000**, the air conditioner **1000** may include two or more displays **1310**.

According to an embodiment of the disclosure, the display **1310** may display an operation mode of the air conditioner **1000**, a current indoor temperature, a current indoor humidity, a current fan speed, and a set temperature (or a desired temperature), but is not limited thereto.

The audio outputter **1320** may output audio data received from the communication interface **1500** or stored in the memory **1800**. For example, the audio outputter **1320** may output an audio signal related to a function performed by the air conditioner **1000** (e.g., notification sound or voice guidance).

According to an embodiment of the disclosure, the outputter **1300** may include an outputter of a remote controller. For example, the operation mode of the air conditioner **1000**, the set temperature, the current fan speed, the current indoor temperature, or the current indoor humidity may be displayed through the outputter of the remote controller. Voice guidance for a user may be output from the remote controller.

According to an embodiment of the disclosure, the outputter **1300** may further include a lighting device (not shown). For example, the outputter **1300** may indicate whether the air conditioner **1000** is operating, an indoor pollution level, or a fan speed by using the lighting device (not shown). The lighting device (not shown) may have various colors.

The outputter **1300** may output information related to replacement of at least one filter included in the filter **1200**. For example, the outputter **1300** may output text, an image (e.g., an icon), or voice for guiding a current state of the filter or a time to replace the filter.

The processor **1400** controls overall operation of the air conditioner **1000**. For example, the processor **1400** may execute programs stored in the memory **1800** to control the blower **1100**, the filter **1200**, the outputter **1300**, the communication interface **1500**, the sensor **1600**, the user inputter **1700**, the memory **1800**, and a power supply (not shown).

According to an embodiment of the disclosure, the processor **1400** may include an AI processor for generating a learning network model, but is not limited thereto. According to an embodiment of the disclosure, the AI processor may be implemented as a chip separate from the processor **1400**. According to an embodiment of the disclosure, the AI processor may be a general-purpose chip.

According to an embodiment of the disclosure, the processor **1400** may execute one or more instructions stored in the memory **1800** to transmit, to the server device **2000**, current state information including at least one of operation time information of the air conditioner **1000**, set temperature information of the air conditioner **1000**, or current indoor temperature information.

The processor **1400** receive a set temperature increase request from the server device **2000** in response to the current state information. When an overcooling period, in which a first predicted temperature graph obtained from the second AI model **202** based on the current state information is reduced below a comfortable temperature graph obtained from the first AI model **201**, is identified, the processor **1400** may receive the set temperature increase request from the server device **2000**. The comfortable temperature graph may include an unstable period in which a comfortable temperature changes based on an operation time of the air conditioner **1000**, and a stable period in which the comfortable temperature is constantly maintained. The comfortable temperature graph may be obtained by applying at least one of operation timing information of the air conditioner **1000** (e.g., an operation time point or time period), fan speed

information of the air conditioner **1000**, indoor humidity information, outdoor humidity information, indoor temperature information, or outdoor temperature information to the first AI model **201**. The first predicted temperature graph may be obtained by further applying at least one of operation timing information of the air conditioner **1000**, indoor humidity information, outdoor temperature information, outdoor humidity information, weather information, performance information of the air conditioner **1000**, or installation space information of the air conditioner **1000** to the second AI model **202**.

The processor **1400** may adjust the set temperature of the air conditioner **1000** based on the set temperature increase request. The set temperature increase request may include information instructing to change the set temperature of the air conditioner **1000** to a first temperature.

According to an embodiment of the disclosure, the processor **1400** may receive a fan speed reduction request from the server device **2000** in response to the current state information. In this case, the current state information may further include fan speed information. The processor **1400** may adjust the fan speed of the air conditioner **1000** based on the fan speed reduction request.

The communication interface **1500** may include one or more elements for enabling communication between the air conditioner **1000** and the server device **2000**, or between the air conditioner **1000** and the display device **3000**. For example, the communication interface **1500** may include a short-range wireless communication interface and a broadcast receiver, but is not limited thereto.

The short-range wireless communication interface may include a Bluetooth communication interface, a BLE communication interface, a NFC interface, a WLAN (or Wi-Fi) communication interface, a Zigbee communication interface, an IrDA communication interface, a WFD communication interface, a UWB communication interface, or an Ant+ communication interface, but is not limited thereto.

The broadcast receiver receives broadcast signals and/or broadcast information from outside through broadcast channels. The broadcast channels may include satellite channels and terrestrial channels. Depending on implementation, the air conditioner **1000** may not include the broadcast receiver.

According to an embodiment of the disclosure, the communication interface **1500** may transmit the current state information to the server device **2000**. The communication interface **1500** may transmit the current state information to the server device **2000** in a certain cycle or at a time requested by the server device **2000**. The current state information may include at least one of set temperature information of the air conditioner **1000** or current indoor temperature information, but is not limited thereto. For example, the current state information may further include fan speed information, indoor humidity information, and device information.

According to an embodiment of the disclosure, the communication interface **1500** may transmit a set temperature adjustment inquiry or a fan speed adjustment inquiry to the server device **2000**. The communication interface **1500** may receive at least one of a set temperature reduction request or a fan speed increase request from the server device **2000**. The communication interface **1500** may receive, from the server device **2000**, information about a timing to inquire about settings adjustment again. For example, the communication interface **1500** may receive, from the server device **2000**, a request instructing to inquire about settings adjustment again after 10 minutes.

According to an embodiment of the disclosure, the communication interface **1500** may obtain location information of the user from a mobile device (not shown) of the user. For example, the communication interface **1500** may obtain the location information (e.g., global positioning system (GPS) information) of the user from the mobile device of the user through short-range wireless communication (e.g., Bluetooth or UWB). According to another embodiment of the disclosure, the communication interface **1500** may obtain the location information of the user from the server device **2000**. In this case, the server device **2000** may be connected to the mobile device of the user and the air conditioner **1000** by using a specific account. For example, when the mobile device of the user uploads location information thereof to the server device **2000**, the communication interface **1500** may receive the location information of the mobile device from the server device **2000**.

The sensor **1600** may include a humidity sensor **1610**, a dust sensor **1620**, and a temperature sensor **1630**, but is not limited thereto. For example, the sensor **1600** may further include a human detection sensor, a gas sensor, a carbon dioxide sensor for measuring a concentration of carbon dioxide in the air, an image sensor for detecting a location of a user, a noise measuring sensor for measuring noise, and a laser sensor for precisely detecting fine particles.

The humidity sensor **1610** may be a sensor for measuring a humidity in the air. The dust sensor **1620** may be a sensor for measuring a concentration of dust in the air. The temperature sensor **1630** may be a sensor for measuring a temperature of the air. The function of each sensor may be intuitively inferred from its name by one of ordinary skill in the art, and thus a detailed description thereof is not provided herein.

The user inputter **1700** refers to the means by which the user inputs data for controlling the air conditioner **1000**. For example, the user inputter **1700** may include a keypad, a dome switch, a touchpad (e.g., a capacitive overlay, resistive overlay, infrared beam, surface acoustic wave, integral strain gauge, or piezoelectric touchpad), a jog wheel, or a jog switch, but is not limited thereto.

According to an embodiment of the disclosure, the user inputter **1700** may include a power button, operation mode buttons (e.g., an AI comfort mode button, a cool mode button, a dry mode button, and a purify mode button), a wind-free function button, a temperature control button, a reservation button, a volume control button, a sleep button, and an auto-sterilize button, but is not limited thereto.

The user inputter **1700** may further include a microphone **1710** for receiving voice input of the user. The microphone **1710** may receive an external audio signal and process the same into electrical voice data. For example, the microphone **1710** may receive an audio signal (e.g., a voice command) from an external device or a user. The microphone **1710** may use various noise removal algorithms to remove noise added while the external audio signal is being received.

According to an embodiment of the disclosure, the user inputter **1700** may include a remote controller and a remote controller receiver **1720**. The remote controller may include a power button, a voice recognition button, an operation mode button, a purify function button, a voice recognition microphone, a wind-free function button, a MAX button, a move and control button, a temperature and fan speed control button, and an additional function select button, but is not limited thereto. According to an embodiment of the disclosure, when the user utters a voice command while pressing the voice recognition button of the remote controller, the remote controller may recognize the voice command

of the user. For example, the user may input a voice command such as 'Turn on the air conditioner', 'Turn off the air conditioner after 1 hour', or 'Operate in AI comfort mode' while pressing the voice recognition button of the remote controller. In this case, the remote controller may provide an answer by voice in response to the voice command of the user. For example, the remote controller may output voice such as "Operating in AI comfort mode. If you feel cold or hot at current temperature, try setting temperature manually".

The remote controller receiver **1720** may receive a control signal from the remote controller. For example, the remote controller receiver **1720** may receive a control signal input by the user, from the remote controller through infrared communication.

The memory **1800** may store programs for processing and control by the processor **1400**, and store input/output data (e.g., operation mode information, user settings information, temperature data, humidity data, gas levels, notifications, filter replacement cycle information, device information, and fan speed information).

The memory **1800** may include at least one type of storage medium from among flash memory, a hard disk, a multimedia card micro, a memory card (e.g., a secure digital (SD) or extreme digital (XD) memory card), random access memory (RAM), static random access memory (SRAM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), programmable read-only memory (PROM), magnetic memory, a magnetic disk, and an optical disc. The programs stored in the memory **1800** may be classified into a plurality of modules depending on functions thereof. The memory **1800** may store at least one AI model.

The air conditioner **1000** may further include the power supply (not shown). The power supply (not shown) may supply power to the elements of the air conditioner **1000** under the control by the processor **1400**. The power supply (not shown) may supply power input from an external power source through a power cord, to each element of the air conditioner **1000** under the control by the processor **1400**.

FIG. **23** is a block diagram for describing functions of the server device **2000**, according to an embodiment of the disclosure.

Referring to FIG. **23**, the server device **2000** may include a communication interface **2100**, a processor **2200**, and a memory **2300**. However, not all of the illustrated elements are required. The server device **2000** may be implemented with more or less elements than the illustrated elements.

The elements of the server device **2000** will now be described one by one.

The communication interface **2100** may include one or more elements for enabling communication between the server device **2000** and the air conditioner **1000**, between the server device **2000** and home appliances, or between the server device **2000** and the display device **3000**.

According to an embodiment of the disclosure, the communication interface **2100** may receive, from the air conditioner **1000**, current state information including at least one of operation time information of the air conditioner **1000**, set temperature information of the air conditioner **1000**, or current indoor temperature information. The communication interface **2100** may receive current humidity information, fan speed information, device information, and user settings information from the air conditioner **1000**.

According to an embodiment of the disclosure, the communication interface **2100** may receive weather information, outdoor humidity information, and outdoor temperature

information from an external server (e.g., a weather server). The communication interface **2100** may obtain indoor humidity information from an air purifier or a humidifier, and obtain, from a cooking appliance (e.g., a gas stove or an induction stove), information about whether the cooking appliance is used. The communication interface **2100** may obtain, from a home network system, information about whether a lighting device producing heat is used, and information about whether windows are open or closed.

The communication interface **2100** may receive location information of a user (e.g., home) from a mobile device of the user. The communication interface **2100** may receive context information of the user from the mobile device of the user. For example, the communication interface **2100** may receive, from the mobile device (e.g., a wearable device) of the user, context information indicating that the user is exercising (e.g., "Exercising"). In this case, the server device **2000** may identify that the user is exercising at home, based on the location information of the user and the context information of the user.

The communication interface **2100** may transmit a set temperature increase request to the air conditioner **1000** in response to the current state information. The communication interface **2100** may transmit a fan speed reduction request to the air conditioner **1000** in response to the current state information. The communication interface **2100** may transmit, to the air conditioner **1000**, information about a timing to inquire again.

The communication interface **2100** may provide information about the air conditioner **1000** to the user through the display device **3000**. For example, the communication interface **2100** may provide information related to an operation mode and energy consumption of the air conditioner **1000** by using a specific application executed by the display device **3000**, but is not limited thereto.

The processor **2200** may control overall operation of the server device **2000** by using programs or information stored in the memory **2300**. According to an embodiment of the disclosure, the processor **2200** may include an AI processor. The AI processor may be produced in the form of a dedicated hardware chip for AI or as a part of a general-purpose processor (e.g., a CPU or an application processor) or a dedicated graphics processor (e.g., a GPU), and be embedded in the server device **2000**.

According to an embodiment of the disclosure, when the current state information is received from the air conditioner **1000**, the processor **2200** may obtain a comfortable temperature graph from the first AI model **201**, and obtain a first predicted temperature graph from the second AI model **202** based on the current state information. The comfortable temperature graph may include an unstable period in which a comfortable temperature changes based on an operation time of the air conditioner **1000**, and a stable period in which the comfortable temperature is constantly maintained. The processor **2200** may obtain the comfortable temperature graph by applying at least one of operation timing information of the air conditioner **1000**, fan speed information of the air conditioner **1000**, indoor humidity information, outdoor humidity information, indoor temperature information, or outdoor temperature information to the first AI model **201**. The processor **2200** may obtain the first predicted temperature graph by further applying at least one of operation timing information of the air conditioner **1000**, indoor humidity information, outdoor temperature information, outdoor humidity information, weather information, performance information of the air conditioner **1000**, or installa-

tion space information of the air conditioner **1000** to the second AI model **202**, in addition to the current state information.

The processor **2200** may compare the comfortable temperature graph to the first predicted temperature graph. The processor **2200** may identify an overcooling period in which the first predicted temperature graph is reduced below the comfortable temperature graph, based on the comparison result. When the overcooling period is identified, the processor **2200** may transmit a set temperature increase request to the air conditioner **1000** through the communication interface **2100**. In this case, the set temperature increase request may include information instructing to change the set temperature of the air conditioner **1000** to a first temperature. The processor **2200** may determine the first temperature from among one or more set temperatures, based on a result of comparing the comfortable temperature graph to one or more second predicted temperature graphs obtained by inputting the one or more set temperatures to the second AI model **202**. A second predicted temperature graph obtained by applying the first temperature to the second AI model **202** may not be reduced below the comfortable temperature graph, and converge on the comfortable temperature graph.

The processor **2200** may generate a fan speed reduction request in response to the current state information. The fan speed reduction request may include information instructing to change the fan speed of the air conditioner **1000** to a first level. The processor **2200** may determine the first level by applying one or more fan speeds to the first and second AI models **201** and **202**.

The memory **2300** may store programs for processing by the processor **2200**, and store input/output data. For example, the memory **2300** may store the first AI model **201**, the second AI model **202**, and the comfortable temperature DB **410**.

The first AI model **201** may be a model for inferring a comfortable temperature of the user. The first AI model **201** may be configured to generate a comfortable temperature graph including an unstable period and a stable period. The first AI model **201** may be configured by training an ANN model (e.g., a linear regression model or a DNN model) based on training data. The first AI model **201** may be refined through training based on updated training data. The training data may include condition information and comfortable temperature information. The condition information may include user settings information (e.g., a desired temperature set by a user and a timing when the user sets the desired temperature), environment information (e.g., an indoor temperature, an indoor humidity, an outdoor temperature, an outdoor humidity, an indoor-outdoor humidity difference, an indoor-outdoor temperature difference, and weather), and device information (e.g., a fan speed currently set in the air conditioner **1000**), but is not limited thereto. For example, the condition information may further include time period information (e.g., morning, afternoon, evening, or dawn), season information, and activity information of the user (e.g., information about whether the user is exercising or cooking). The comfortable temperature information may include a temperature at which the user feels comfortable under each condition.

The first AI model **201** may obtain a comfortable temperature graph as an output value based on at least one of time information, environment information, or device information. The time information may include operation timing information of the air conditioner **1000** (e.g., an operation time point or time period). The environment information

may include an indoor humidity, an outdoor humidity, an indoor temperature, an outdoor temperature, an indoor-outdoor temperature difference, an indoor-outdoor humidity difference, and weather. The device information may include fan speed information, device performance information, and device decrepitude information. The first AI model **201** may extract the comfortable temperature graph from the comfortable temperature DB **410**, based on the time information, the environment information, and the device information.

The second AI model **202** may be a model for predicting indoor temperatures of future timings. The second AI model **202** may be configured by training an ANN model (e.g., a DNN model, an RNN model, or a CNN model) based on training data. The training data may include user settings information (e.g., a desired temperature set by a user and a timing when the user sets the desired temperature), environment information (e.g., an indoor temperature, an indoor humidity, an outdoor temperature, an outdoor humidity, an indoor-outdoor humidity difference, an indoor-outdoor temperature difference, and weather), device information (e.g., a fan speed currently set in the air conditioner **1000**, and device performance), and space information (e.g., thermal insulation information, information about whether a lighting device is used, and information about whether windows are open or closed).

The second AI model **202** may obtain a predicted temperature graph as an output value based on time information, environment information, device information, and space information. According to an embodiment of the disclosure, not all of the time information, the environment information, the device information, and the space information may be applied to the second AI model **202**, and only some of them may be applied. The time information may include operation time information of the air conditioner **1000**, and operation timing information of the air conditioner **1000** (e.g., an operation time point or time period). The environment information may include an indoor humidity, an outdoor humidity, an indoor temperature, an outdoor temperature, an indoor-outdoor temperature difference, an indoor-outdoor humidity difference, and weather. The device information may include fan speed information set in the air conditioner **1000**, set temperature information of the air conditioner **1000**, device performance information, and device decrepitude information, but is not limited thereto. The space information may include information about a thermal insulation level, information about whether a cooking appliance producing heat (e.g., a gas stove or an induction stove) is used, information about whether a lighting device producing heat is used, and information about whether windows are open or closed, but is not limited thereto. The predicted temperature graph may be generated in various forms based on a set temperature of the air conditioner **1000**, a fan speed, an operation time of the air conditioner **1000**, and a current indoor temperature.

The comfortable temperature DB **410** may be a DB storing comfortable temperature graphs of the user under various conditions, and include a table in which the condition information is mapped to a comfortable temperature graph of the user. The condition information may include time information (e.g., operation timing information of the air conditioner **1000**), environment information (e.g., an indoor humidity, an outdoor humidity, an indoor temperature, an outdoor temperature, an indoor-outdoor humidity difference, an indoor-outdoor temperature difference, and weather), and device information (e.g., fan speed information and performance information), but is not limited thereto.

The first AI model **201** may be trained based on training data to generate the comfortable temperature DB **410**. According to an embodiment of the disclosure, the first AI model **201** may update the comfortable temperature DB **410** when the training data is added or updated.

According to an embodiment of the disclosure, the air conditioner **1000** may adjust a set temperature and/or a fan speed by using the first AI model **201** driven by the server device **2000** to infer a comfortable temperature and the second AI model **202** driven by the server device **2000** to predict an indoor temperature, in such a manner that a user does not experience discomfort due to overcooling.

The method according to an embodiment of the disclosure may be implemented in the form of program commands executable by various computer means and be recorded on a computer-readable medium. The computer-readable medium may include program commands, data files, data structures, or a combination thereof. The program commands recorded on the medium may be those specially designed and configured for the disclosure, or those known to and usable by one of ordinary skill in the art of computer software. Examples of the computer-readable recording medium include magnetic media (e.g., hard disks, floppy disks, and magnetic tape), optical media (e.g., CD-ROMs and DVDs), magneto-optical media (e.g., floptical disks), and hardware devices (e.g., ROMs, RAMs, and flash memories) that are specially configured to store and execute program commands. Examples of the program commands include both machine code, such as produced by a compiler, and high-level language code that may be executed by the computer using an interpreter.

An embodiment of the disclosure may be implemented in the form of a recording medium including computer-executable instructions, e.g., program modules executed by the computer. The computer-readable medium may be an arbitrary available medium that can be accessed by the computer, and examples thereof include all of volatile, non-volatile, detachable, and non-detachable media. The computer-readable medium may include both of a computer storage medium and a communication medium. Examples of the computer storage medium include all of volatile, non-volatile, detachable, and non-detachable media implemented using an arbitrary method or technology for storing information such as computer-readable instructions, data structures, program modules, or other data. The communication medium typically includes computer-readable instructions, data structures, program modules, other data in modulated data signals such as carrier waves, or other transmission mechanisms, and includes an arbitrary information delivery medium. An embodiment of the disclosure may also be implemented in the form of a computer program or a computer program product including computer-executable instructions, e.g., computer programs executed by the computer.

A machine-readable storage medium may be provided in the form of a non-transitory storage medium. When the storage medium is 'non-transitory', it means that the storage medium is tangible and does not include signals (e.g., electromagnetic waves), and it does not limit that data is semi-permanently or temporarily stored in the storage medium. For example, the 'non-transitory storage medium' may include a buffer for temporarily storing data.

According to an embodiment of the disclosure, the method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between sellers and purchasers. The computer pro-

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gram product may be distributed in the form of a machine-readable storage medium (e.g., a compact disc read only memory (CD-ROM)), or be electronically distributed (e.g., downloaded or uploaded) via an application store or directly between two user devices (e.g., smartphones). For electronic distribution, at least a part of the computer program product (e.g., a downloadable app) may be temporarily generated or be at least temporarily stored in a machine-readable storage medium, e.g., a server of a manufacturer, a server of an application store, or a memory of a relay server.

While the disclosure has been particularly shown and described with reference to embodiments of the disclosure, it will be understood by one of ordinary skill in the art that various changes in form and details may be made therein without departing from the scope of the disclosure as defined by the following claims.

What is claimed is:

1. An air conditioner comprising:

a communication interface to communicate with a server device;

a memory to store one or more instructions; and

at least one processor configured to execute the stored one or more instructions to:

transmit, to the server device through the communication interface, current state information comprising one or more of operation time information of the air conditioner, set temperature information of the air conditioner, and current indoor temperature information;

receive a set temperature increase request corresponding to the transmitted current state information from the server device through the communication interface; and

adjust a set temperature of the air conditioner based on the received set temperature increase request,

wherein a comfortable temperature graph is obtained from a first Artificial Intelligence (AI) model and a first predicted temperature graph is obtained from a second AI model based on the transmitted current state information,

wherein the set temperature increase request is received when an overcooling period, in which the obtained first predicted temperature graph is reduced below the obtained comfortable temperature graph, is identified, and

wherein the comfortable temperature graph comprises an unstable period in which a comfortable temperature changes based on an operation time of the air conditioner, and a stable period in which the comfortable temperature is constantly maintained.

2. The air conditioner of claim 1, wherein the comfortable temperature graph is obtained by applying one or more of operation timing information of the air conditioner, fan speed information of the air conditioner, indoor humidity information, outdoor humidity information, indoor temperature information, and outdoor temperature information to the first AI model.

3. The air conditioner of claim 1, wherein the set temperature increase request comprises information instructing to change the set temperature of the air conditioner to a first temperature, and

wherein the first temperature is determined from among one or more set temperatures, based on a result of comparing the comfortable temperature graph to one or more second predicted temperature graphs obtained by inputting the one or more set temperatures to the second AI model.

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4. The air conditioner of claim 3, wherein a second predicted temperature graph obtained by applying the first temperature to the second AI model is not reduced below the comfortable temperature graph and is closest to the comfortable temperature graph.

5. The air conditioner of claim 1, wherein the processor is further configured to execute the one or more instructions to: receive a fan speed reduction request from the server device corresponding to the transmitted current state information; and

adjust a fan speed of the air conditioner based on the fan speed reduction request.

6. The air conditioner of claim 1, wherein the first predicted temperature graph is obtained by further applying one or more of operation timing information of the air conditioner, indoor humidity information, outdoor temperature information, outdoor humidity information, weather information, performance information of the air conditioner, and installation space information of the air conditioner to the second AI model.

7. An operating method of an air conditioner, the operating method comprising:

transmitting, to a server device, current state information comprising one or more of operation time information of the air conditioner, set temperature information of the air conditioner, and current indoor temperature information;

receiving a set temperature increase request corresponding to the transmitted current state information from the server device; and

adjusting a set temperature of the air conditioner based on the received set temperature increase request,

wherein a comfortable temperature graph is obtained from a first Artificial Intelligence (AI) model and a first predicted temperature graph is obtained from a second AI model based on the transmitted current state information,

wherein the set temperature increase request is received when an overcooling period, in which the obtained first predicted temperature graph is reduced below the obtained comfortable temperature graph, is identified, and

wherein the comfortable temperature graph comprises an unstable period in which a comfortable temperature changes based on an operation time of the air conditioner, and a stable period in which the comfortable temperature is constantly maintained.

8. The operating method of claim 7, wherein the comfortable temperature graph is obtained by applying one or more of operation timing information of the air conditioner, fan speed information of the air conditioner, indoor humidity information, outdoor humidity information, indoor temperature information, and outdoor temperature information to the first AI model.

9. The operating method of claim 7, wherein the set temperature increase request comprises information instructing to change the set temperature of the air conditioner to a first temperature, and

wherein the first temperature is determined from among one or more set temperatures, based on a result of comparing the comfortable temperature graph to one or more second predicted temperature graphs obtained by inputting the one or more set temperatures to the second AI model.

10. The operating method of claim 9, wherein a second predicted temperature graph obtained by applying the first

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temperature to the second AI model is not reduced below the comfortable temperature graph and is closest to the comfortable temperature graph.

11. The operating method of claim 7, further comprising: receiving a fan speed reduction request from the server device corresponding to the transmitted current state information; and adjusting a fan speed of the air conditioner based on the fan speed reduction request.

12. The operating method of claim 7, wherein the first predicted temperature graph is obtained by further applying one or more of operation timing information of the air conditioner, indoor humidity information, outdoor temperature information, outdoor humidity information, weather information, performance information of the air conditioner, and installation space information of the air conditioner to the second AI model.

13. A control system comprising:

an air conditioner configured to transmit, to a server device, current state information comprising one or more of operation time information of the air conditioner, set temperature information of the air conditioner, and current indoor temperature information, and adjust a set temperature of the air conditioner based on a set temperature increase request, corresponding to the transmitted current state information, received from the server device; and

the server device configured to transmit the set temperature increase request to the air conditioner when an overcooling period, is identified,

wherein a comfortable temperature graph is obtained from a first Artificial Intelligence (AI) model and a first predicted temperature graph is obtained from a second AI model based on the transmitted current state information,

wherein the overcooling period is a period when the obtained first predicted temperature is reduced below the obtained comfortable temperature graph, and

wherein the comfortable temperature graph comprises an unstable period in which a comfortable temperature

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changes based on an operation time of the air conditioner, and a stable period in which the comfortable temperature is constantly maintained, and is obtained by applying one or more of operation timing information of the air conditioner, fan speed information of the air conditioner, indoor humidity information, outdoor humidity information, indoor temperature information, and outdoor temperature information to the first AI model.

14. The control system of claim 13, wherein the set temperature increase request comprises information instructing to change the set temperature of the air conditioner to a first temperature, and

wherein the server device is further configured to determine the first temperature from among one or more set temperatures, based on a result of comparing the comfortable temperature graph to one or more second predicted temperature graphs obtained by inputting the one or more set temperatures to the second AI model.

15. The control system of claim 14, wherein a second predicted temperature graph obtained by applying the first temperature to the second AI model is not reduced below the comfortable temperature graph and is closest to the comfortable temperature graph.

16. The control system of claim 13, wherein the air conditioner is further configured to receive a fan speed reduction request from the server device corresponding to the transmitted current state information, and adjust a fan speed of the air conditioner based on the fan speed reduction request.

17. The control system of claim 13, wherein the server device is further configured to obtain the first predicted temperature graph from the second AI model by further applying one or more of operation timing information of the air conditioner, indoor humidity information, outdoor temperature information, outdoor humidity information, weather information, performance information of the air conditioner, and installation space information of the air conditioner to the second AI model.

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